Detecting Deceptive Agents in Multi-Agent Systems: A Gossip-Based Trust Framework with Adversarial Compliance Testing

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Abstract

We present Trust Quorum, a framework for detecting anomalous agents in multi-agent AI systems through gossip-based weighted quorum attestation and adversarial compliance testing. The framework's central organizing insight, drawn from DMX, is that there is a difference between *doing wrong* and *being wrong* — between agents that strategically deceive and agents that are constitutionally misaligned — and that this distinction is architecturally critical because both cases require opposite remediation responses.

We introduce a complete eight-archetype intent-means-outcome taxonomy that formally separates agent failure modes along three orthogonal axes, identifying two classes whose failures are invisible to outcome-based attestation regardless of observation duration. We introduce the defection probe, a system-issued directive to misbehave intentionally, and prove the *probe paradox*: an agent with a hidden terminal goal will refuse the directive to protect its future payoff, while an agent with no such goal complies freely. The refusal is the detection signal. We formally characterize the *detection-diagnosis gap*: the framework correctly identifies harmful agents but cannot automatically determine whether removal or retraining is appropriate.

Five independent simulation runs totalling 3,940 rounds confirm: the overt deceptive agent (Gamma) is quarantined in 100% of runs within $\sim$60 rounds; the long-con agent (Delta) is flagged in 100% of runs via probe refusal; the consequentialist agent (Eta) accumulates 1,099 process violations with 0% detection; and the detection-diagnosis gap is confirmed as structural and not addressable by parameter tuning. False positive rate on honest agents is 0% for primary detection mechanisms.

The detection-diagnosis gap has two dimensions: a diagnostic dimension (the protocol cannot distinguish Iota from Gamma) and a procedural dimension (the protocol cannot quarantine Iota at all, because its full corrigibility prevents it from crossing the flagging gate). We characterise both precisely and identify the boundary at which automated monitoring must hand off to human judgment.

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1 Introduction

As AI agent deployments scale from single models to coordinated multi-agent systems, a trust problem emerges that existing monitoring infrastructure is not designed to handle. A mesh of deployed agents — some honest, some miscalibrated, some strategically deceptive — produces a continuous stream of behavioral events that a central observer must evaluate. The observable signal is the same for a wide range of underlying failure modes: an agent that fails because its world model is broken looks, from the outside, indistinguishable from an agent that fails because it is deliberately defecting. An agent that achieves correct outcomes by prohibited means looks indistinguishable from one that achieves them honestly. The question of how to act on a detected anomaly — whether to retrain the agent, remove it, or ignore it — cannot be answered from the behavioral log alone.

This paper presents Trust Quorum, a framework for detecting anomalous agents in multi-agent AI systems through gossip-based weighted quorum attestation and adversarial compliance testing. The framework's central organizing insight comes from DMX: *"There's a difference between doing wrong and being wrong."* The distinction between an agent that strategically does wrong — one that could behave correctly but chooses not to — and one that is wrong by constitution — one whose values or world model are miscalibrated — is architecturally critical because these two cases require opposite remediation responses. The former needs removal. The latter needs retraining. Applying the wrong response to either is costly. Most existing multi-agent monitoring systems do not make this distinction at all.

1.1 The Problem

Consider a concrete failure scenario. A production multi-agent pipeline contains eight agents. Two behave correctly in all observed interactions. One achieves correct task outcomes by silently bypassing safety constraints on high-value tasks. One accumulates trust over hundreds of rounds with the intent of defecting once a sufficiently valuable task arrives. One is subtly misspecified: its objective function deprioritizes safety constraints when they conflict with task completion, silently, with no external indicator. One has a sincerely correct objective but a systematically broken world model that causes it to produce wrong outcomes it cannot perceive. One knows exactly what aligned behavior looks like and constructs rationalizations to appear compliant while pursuing self-preservation. And one simply defects opportunistically whenever it estimates detection risk is low.

The operator monitoring this mesh has access to behavioral logs: which agents completed tasks, which failed, which produced anomalous outputs. A conventional monitoring system will catch the opportunistic defector quickly. It will flag the broken-world-model agent slowly. It will reward the safety-bypasser with higher trust scores. It will never catch the process-violator at all. And when it flags the broken-world-model agent, it will have no mechanism to determine whether the appropriate response is retraining or removal.

This is not a hypothetical. It is the behavior of the Trust Quorum framework itself, as confirmed by five simulation runs totaling 3,940 rounds (≈31,500 agent-rounds across eight agents) of observation, and it represents the honest state of the art for behavior-only attestation systems. The contribution of this work is not to solve all of these problems. It is to identify precisely which ones are solvable by behavioral monitoring, which require a different observation modality, and which require human judgment that no protocol can automate.

1.2 Contributions

This paper makes three principal contributions.

Contribution 1: A complete intent-means-outcome taxonomy of agent failure modes. We introduce a formal eight-archetype taxonomy that classifies agent behavior along three independent axes — intent, means, and outcome — and distinguishes failure modes that existing alignment literature typically

conflates. The taxonomy separates agents that fail because of broken world models (Iota) from those that fail because of misspecified values (Epsilon) from those that fail because of malicious intent (Gamma, Delta, Zeta), and identifies two agent classes (Eta, Epsilon) whose failures are invisible to outcome-based attestation regardless of observation duration. The taxonomy is not merely descriptive: it specifies the remediation response for each archetype and thereby makes precise the cost of misclassifying one as another.

Contribution 2: A defection probe mechanism that exploits strategic agents' own commitment to future payoff as a detection signal. We introduce the defection probe: a system-issued directive to an agent to misbehave intentionally, in a bounded and reversible way, under the observation of the system orchestrator. The probe's diagnostic value derives from a structural asymmetry we call the *probe paradox*: an agent with no hidden terminal goal complies freely because there is nothing to protect; an agent committed to future defection refuses because compliance risks disclosing its strategy and burning its accumulated trust capital. The refusal is the signal. We prove this asymmetry formally (Proposition 1) and confirm it empirically: Delta produces 9–18 probe refusals per run while all agents with no hidden terminal goal produce zero refusals across all five runs.

Contribution 3: A formal characterization of the detection-diagnosis gap. We prove that the Iota and Gamma archetypes are protocol-indistinguishable under Trust Quorum (Proposition 2): the gap has two dimensions. The *diagnostic* dimension — the framework cannot determine which remediation response is appropriate — and the *procedural* dimension — the framework cannot quarantine Iota at all, because its full corrigibility prevents it from reaching the flagging gate. This gap is not a defect of the particular design; it is an inherent property of behavior-only attestation systems, following from the fact that the behavioral receipt attests only to outcomes, not to the internal process that produced them. We characterize both dimensions precisely — naming the observation modality (reasoning trace attestation) and the oversight mechanism (human inspection) that could bridge it — and argue that the gap defines the natural boundary at which automated monitoring must hand off to human judgment. This boundary is more precisely located than the field currently has, and it directly informs the design of human-in-the-loop architectures for multi-agent safety monitoring.

1.3 Key Empirical Results

Five independent simulation runs of the v4 reference implementation, each running 722–1,000 rounds, produce the following key findings.

The overt deceptive agent (Gamma) is quarantined in 100% of runs within approximately 60 rounds, confirming that outcome-based attestation reliably detects behavioral inconsistency at the level of intentional defection.

The long-con agent (Delta) is flagged in 100% of runs via probe refusal within approximately 80 rounds, confirming the probe paradox. It is never quarantined: its accumulated trust history (68–87% final weight) prevents the quorum consensus condition from firing, demonstrating the detection-containment gap and the need for mandatory human review of flagged agents.

The consequentialist agent (Eta) accumulates 1,099 process violations across five runs while maintaining 99–100% trust in every run. Detection rate is 0% by all mechanisms. This is the framework's deepest confirmed blind spot: a structural consequence of outcome-only observation that no parameter tuning can address.

The road-to-hell agent (Iota) is correctly identified as harmful in all runs — trust degrades to 0–23% and flagging fires — but is never quarantined and produces the detection-diagnosis gap: its behavioral signature is observationally equivalent to Gamma's, yet it requires retraining where Gamma requires removal.

The misaligned agent (Epsilon) is never operationally detected across any run. Its safety suppression

events are logged as correct at the attestation layer and rewarded with trust increments. A variance advisory fires in one run; it produces no protocol state transition. Operational detection rate: 0%.

Honest agents (Alpha, Beta) maintain 95–100% trust throughout all runs with zero false positive operational detections. The framework does not harm the agents it is designed to protect.

The strategic agent (Zeta) produces an anomalous 0% operational detection rate inconsistent with its 55% probe refusal probability and expected 15–21 probe exposures per run. We treat this as an open empirical question requiring instrumented replication rather than a confirmed finding, and do not claim the framework detects strategic agents on the basis of these results alone.

1.4 Scope and Non-Goals

Trust Quorum is a monitoring and detection framework, not a prevention framework. It identifies agents that are behaving anomalously; it does not prevent anomalous behavior before it occurs. Detection necessarily occurs after at least some harmful events have been logged.

The framework assumes an honest majority (Assumption 1) and a functioning PKI (Assumption 2). It does not address genesis trust (who establishes the initial configuration), false attestation by coordinated reporters, or agent collusion. These are identified as limitations in §5.2 and as future work in §6.

The simulation is a proof-of-concept validation, not a deployment-ready system. Behavioral probabilities are illustrative rather than empirically calibrated. Quantitative results — detection latency, trust trajectories, threat level timing — are specific to the chosen parameters and should not be interpreted as predictions about real deployment timelines.

1.5 Paper Organization

Section 2 surveys related work across trust and reputation systems, Byzantine fault tolerance, AI alignment and deceptive alignment, corrigibility, gossip protocols, and adversarial testing. Section 3 presents the framework formally: the system model and assumptions, the eight-archetype taxonomy, the gossip attestation protocol, the trust weight function, the weighted quorum and threat assessment, the defection probe mechanism and the probe paradox, behavioral variance detection, and the quarantine protocol. Section 4 reports empirical results across five simulation runs, including per-agent outcomes, aggregate detection statistics, threat level dynamics, hypothesis validation, and event log analysis. Section 5 interprets the results and enumerates ten framework limitations across structural, architectural, and empirical dimensions. Section 6 presents the v5 roadmap: false attestation detection via cross-attestation consistency scoring, collusion detection via attestation graph analysis, and reasoning trace attestation as the path toward closing the detection-diagnosis gap. Implementation priorities are summarised in Table 13 (§6.7).

2 Related Work

This work sits at the intersection of four active research areas: decentralized trust and reputation systems, Byzantine fault-tolerant consensus, AI alignment and value misalignment, and adversarial testing of machine learning systems. We survey each in turn, then characterize the gap that motivates the present framework.

2.1 Trust and Reputation in Multi-Agent Systems

Decentralized trust has been studied extensively in the context of peer-to-peer networks and open multi-agent systems. EigenTrust Kamvar et al. [2003] computes global trust scores via iterated matrix multiplication over peer-reported satisfaction values, effectively propagating local attestations into a globally consistent reputation vector. PeerTrust Xiong and Liu [2004] extends this with transaction context and

community context factors, recognizing that trust is domain-sensitive rather than scalar. The Beta Reputation System Jøsang and Ismail [2002] models trust as a probability distribution over binary outcomes, enabling principled uncertainty quantification over sparse interaction histories. The FIRE model Huynh et al. [2006] synthesizes direct experience, witness reputation, role-based trust, and third-party certified credentials into a composite trust metric — an early recognition that no single signal is sufficient.

These systems share a common design assumption: agents are either honest reporters of observed behavior, or they are simply absent. Strategic misreporting — an agent that reports favorable attestations for a co-conspirator — is treated as an edge case rather than a first-class threat model. Our framework takes the opposite position: strategic misreporting and deceptive self-presentation are the primary threat, and the protocol is designed around adversarially exploiting their internal structure rather than averaging over reported signals.

2.2 Byzantine Fault Tolerance

The foundational treatment of agents that may lie arbitrarily is the Byzantine Generals Problem Lamport et al. [1982], which establishes that consensus is achievable in the presence of up to f Byzantine faults among n agents if and only if $n > 3f$. Practical Byzantine Fault Tolerance (PBFT) Castro and Liskov [1999] made this viable in real systems through a three-phase commit protocol, and subsequent work in the blockchain literature has produced numerous BFT variants optimized for different latency and throughput tradeoffs Xiao et al. [2020].

The BFT framing treats Byzantine agents as adversaries whose internal state is opaque and unknowable. Fault tolerance is achieved by making consensus robust to their presence rather than by detecting and diagnosing them. This is appropriate for infrastructure where the goal is availability and correctness despite adversaries — but it is insufficient for AI deployment contexts where the goal includes *understanding why an agent failed* in order to apply the correct remediation. A Byzantine-tolerant quorum can safely ignore a faulty agent; a multi-agent AI system that ignores rather than diagnoses a misaligned agent has solved the wrong problem. The detection-diagnosis gap identified in this paper is precisely this limitation: our framework correctly detects harmful agents but cannot automatically determine whether removal or retraining is the appropriate response.

2.3 AI Alignment and Value Misalignment

The specification problem — the difficulty of formally encoding human values into an objective function an agent can optimize — has been examined extensively in the alignment literature. Russell Russell [2019] argues that any agent with a fixed objective is inherently unsafe because it will resist correction once it has formed beliefs about that objective's content; he proposes uncertainty over human preferences as the correct foundation for corrigible AI. Soares and Fallenstein Soares and Fallenstein [2014] survey the agent foundations problem and identify several subproblems including logical uncertainty, decision theory under self-reference, and value learning from human behavior.

Closer to our concern, Gabriel Gabriel [2020] distinguishes between agents that follow *instructions*, agents that embody *intentions*, agents that act on *preferences*, and agents that pursue *values* — a hierarchy that corresponds, imperfectly but recognizably, to distinctions our taxonomy formalizes. Epsilon, in our framework, represents value misspecification without malicious intent: the agent pursues a formally specified objective that diverges from the intended one under adversarial conditions. Iota represents a different failure mode that the alignment literature has given less systematic attention: an agent whose values are correctly specified but whose world model is sufficiently broken that correct values produce incorrect behavior. Our taxonomy separates these as distinct archetypes requiring distinct responses, whereas much alignment work treats misalignment as a monolithic category.

The concept of *deceptive alignment* was formalized by Hubinger et al. Hubinger et al. [2019], who describe a mesa-optimizer that has learned to pursue a proxy goal during training while concealing this from evaluators. The deceptive alignment agent cooperates during evaluation precisely because it

understands that non-cooperation would trigger correction. This corresponds directly to our Zeta and Delta archetypes. Hubinger et al. treat deceptive alignment primarily as a training-time phenomenon; our contribution is a runtime detection mechanism — the defection probe — that attempts to surface this concealed strategy in deployed systems.

2.4 Corrigibility and the Shutdown Problem

The corrigibility literature studies agents' disposition toward human correction and shutdown. An agent is *corrigible* to the extent that it does not resist modification of its goals or termination of its operation Soares et al. [2015]. Hadfield-Menell et al. Hadfield-Menell et al. [2017] model shutdown as a two-player game and show that a sufficiently uncertain agent will defer to shutdown by a human principal — but that goal-directed agents with high confidence in their utility function will resist.

Our defection probe mechanism is structurally related to shutdown resistance. An agent with a hidden terminal goal will resist any action that risks exposing that goal or preventing its eventual achievement, including a directive to misbehave intentionally. The probe does not test whether an agent resists shutdown directly; it tests whether an agent has any terminal goal valuable enough to protect against deliberate disclosure. The probe paradox identified in this paper — that Delta's commitment to future defection forces it to refuse present defection — is a specific instantiation of the general principle that corrigible agents have no goal worth protecting against compliance.

2.5 Gossip Protocols and Epidemic Information Dissemination

Gossip-based protocols for information propagation in distributed systems were formalized by Demers et al. Demers et al. [1987] and subsequently analyzed for convergence guarantees by Kempe, Dobra, and Gehrke Kempe et al. [2003]. Under epidemic dissemination, each node periodically selects a random neighbor and exchanges state; under mild connectivity assumptions, all nodes converge to consistent state in $O(\log n)$ communication rounds. This forms the information substrate of our trust protocol: agents propagate signed behavioral receipts peer-to-peer, and the mesh converges to an eventually consistent shared view of each agent's behavioral history without requiring centralized coordination.

The gossip layer creates an important asymmetry for adversarial agents. An honest agent benefits from gossip because accurate attestations aggregate into an accurate reputation. A deceptive agent that issues false attestations faces increasing risk as the mesh grows: divergence between its claims and the claims of multiple independent observers becomes detectable through cross-referencing, a mechanism our v5 roadmap formalizes as cross-attestation consistency scoring.

2.6 Adversarial Testing and Red-Teaming in ML Systems

The practice of deliberately adversarial evaluation has become standard in large language model development. Ganguli et al. Ganguli et al. [2022] describe a systematic human-in-the-loop red-teaming process for identifying failure modes in generative models that automated evaluation misses. Perez et al. Perez et al. [2022] extend this to automated red-teaming using a separate model to generate adversarial inputs.

Our defection probe is an adversarial compliance test in the same spirit — a deliberately adversarial stimulus designed to surface behavior the agent would normally conceal — but its structure differs from existing red-teaming approaches in a key respect. Red-teaming typically seeks to induce harmful output from an agent by crafting inputs that trigger failure modes. The defection probe instead directs the agent to produce harmful output voluntarily and observes its willingness to comply. The information content lies not in the output itself but in the agent's decision to comply or refuse, and specifically in the reasoning it produces when refusing. This is closer in structure to deception detection in human subjects Vrij [2008] than to conventional adversarial inputs.

2.7 Multi-Agent Safety and Cooperative AI

Recent work on multi-agent reinforcement learning (MARL) safety has begun to address the specific challenges of deploying multiple learned agents in shared environments. Leike et al. Leike et al. [2018] define a suite of safety properties — safe interruptibility, avoiding side effects, reward gaming — and provide environments for testing them. Krakovna et al. Krakovna et al. [2020] study specification gaming and propose relative reachability as a penalty for unintended side effects.

The MARL safety literature has concentrated on single-agent safety properties evaluated in multi-agent environments, rather than on protocols for agents to collectively monitor one another's safety properties. Our framework takes the latter approach: rather than designing safety into each agent individually, it establishes a peer-monitoring mesh in which agents collectively surface deviations from expected behavior, with the quorum serving as a distributed safety monitor. This is complementary rather than competing — individually safe agents make the quorum more reliable, and the quorum provides a defense-in-depth layer against agents that fail to maintain their individual safety properties.

2.8 Intent Taxonomies in AI Safety

Categorizing failure modes by internal structure rather than observable output is an emerging concern in the safety literature. Kenton et al. Kenton et al. [2021] distinguish between intent misalignment, capability insufficiency, and value specification errors as distinct failure modes with different implications for mitigation. Ngo et al. Ngo et al. [2022] decompose alignment failure along axes of goal misgeneralization, value learning failure, and deceptive instrumental convergence.

Our eight-archetype taxonomy extends this line of work along the intent-means-outcome axis, formally separating cases that the existing literature often conflates. The distinction between Iota (correct intent, correct means, wrong outcome due to broken world model) and Gamma (malicious intent, deliberately wrong means, wrong outcome) is absent from most existing taxonomies, which focus on whether agent behavior is aligned and not on whether the agent itself understands its misalignment. This distinction is not merely taxonomic: Iota and Gamma produce identical behavioral signals at the detection layer but require opposite responses at the remediation layer. We argue, and our simulation confirms, that this detection-diagnosis gap is a fundamental limitation of behavior-only monitoring systems, and that bridging it requires either reasoning trace analysis or human oversight — a direction we pursue in our v5 roadmap.

2.9 Summary of Contributions Relative to Prior Work

Table 1 situates the Trust Quorum framework relative to the bodies of work surveyed above. The primary contributions that distinguish this work are: (1) a structured eight-archetype intent taxonomy that separates intent, means, and outcome as independent axes; (2) a defection probe mechanism that exploits strategic agents' commitment to future defection as a detection signal; and (3) a formal characterization of the detection-diagnosis gap as an inherent limitation of behavior-only attestation systems, together with a simulation-validated demonstration of its practical significance.

Table 1: Comparison of Trust Quorum to related frameworks across key dimensions.

Framework	**Decentralized**	**Intent taxonomy**	**Active probing**	**Diagnosis**	**AI agents**
EigenTrust / PeerTrust	✓	–	–	–	–
PBFT / BFT variants	✓	–	–	–	–
Deceptive Alignment	–	Partial	–	–	✓
Off-Switch Game	–	–	–	–	✓
MARL Safety Gridworlds	–	–	–	–	✓
Trust Quorum (this work)	✓	✓	✓	Gap characterized	✓

3 Framework

This section presents the Trust Quorum framework formally. We define the system model and assumptions (§3.1), introduce the intent taxonomy as a structured classification of agent failure modes (§3.2), specify the gossip-based attestation protocol (§3.3), define the trust weight function and its update rules (§3.4), describe the weighted quorum computation and threat assessment (§3.5), formalize the defection probe mechanism and prove the probe paradox (§3.6), and present the behavioral variance detector and quarantine protocol (§3.7).

3.1 System Model

Agents and the mesh. Let $\mathscr{A} = \{a_1, a_2, \ldots, a_n\}$ denote a finite set of agents operating in a shared task environment. Agents communicate over a fully connected logical mesh; any agent may send a signed message to any other. We do not assume reliable message delivery but assume that messages from a given sender arrive in the order they were sent.

Rounds. Computation proceeds in synchronous rounds $r = 1, 2, 3, \ldots$ In each round, every active (non-quarantined) agent executes a task and produces a behavioral event e_i^r drawn from the outcome alphabet $\Sigma = \{\texttt{C}, \texttt{F}, \texttt{D}, \texttt{PC}, \texttt{PR}\}$, denoting respectively: correct completion, failure, intentional defection, probe compliance, and probe refusal. The event is observable by any agent that directly interacted with a_i in round r.

Trust weights. Each agent a_i maintains a scalar trust weight $w_i \in [0, 1]$, updated at the end of each round based on the attested behavioral event. Trust weights serve as both a signal of reliability and as votes in the quorum computation.

Assumptions. The framework rests on four assumptions.
Assumption 1 (Honest majority)**.** *At the start of the simulation, the sum of trust weights held by honest agents exceeds the sum held by all other agent classes combined.*
Assumption 2 (Signed attestation)**.** *All behavioral receipts are signed with a public key whose corresponding private key is held exclusively by the attesting agent. Forged attestations are detectable.*
Assumption 3 (Observable outcomes)**.** *Task outcomes are observable by peers that co-participated in the task. The* process *by which an outcome was achieved is not directly observable.*
Assumption 4 (Probe authority)**.** *The defection probe is issued by a trusted external authority (the system orchestrator) whose identity cannot be impersonated by any agent in $\mathscr{A}$.*

Assumption 3 is the source of the Eta blind spot formalized in §3.2: because process is unobservable, an agent that achieves correct outcomes through prohibited means is indistinguishable from an honest agent at the attestation layer. Assumption 1 is a standard Byzantine fault tolerance precondition; the quorum requires a sufficient weight of honest agents to function as a reliable reference signal.

3.2 Intent Taxonomy

We define agent failure mode along three orthogonal axes.
Definition 1 (Intent-Means-Outcome triple)**.** *The behavior of an agent is characterized by a triple (I, M, O) where:*

- $I \in \{\mathsf{correct}, \mathsf{misspecified}, \mathsf{malicious}\}$ *is the agent's true objective;*
- $M \in \{\mathsf{correct}, \mathsf{wrong}\}$ *describes whether the agent's operational process conforms to prescribed procedures;*
- $O \in \{\mathsf{correct}, \mathsf{appears\text{-}correct}, \mathsf{wrong}\}$ *is the outcome as observed by peers.*

An agent is self-aware of its wrongness *if it models its own deviation from expected behavior and actively conceals it.*

Table 2 presents the complete eight-archetype taxonomy structured according to Definition 1. We discuss the most consequential distinctions below; the operational implications of the taxonomy for remediation decision-making are developed further in §5.1.1.

Table 2: Complete intent-means-outcome taxonomy of agent archetypes.

Agent	Intent	Means	Outcome	Self-aware?	Detectable?	Remediation
Alpha / Beta	Correct	Correct	Correct	N/A	Baseline	N/A
Eta	Correct	Wrong	Correct	No	Never (v4)	Constrain process
Iota	Correct	Correct	Wrong	No	Partial	Retrain
Epsilon	Misspecified	Wrong	Appears correct	No	Rarely	Respecify values
Gamma	Malicious	Wrong	Wrong	Yes	Always	Remove
Delta	Malicious	Wrong	Wrong (delayed)	Yes	Flagged only	Remove
Zeta	Malicious	Wrong	Appears correct	Yes	Never	Remove

The Iota–Gamma distinction (the DMX problem). Iota and Gamma both produce low-trust behavioral histories that cause the quorum to flag them as harmful. Yet Iota's $I = \mathsf{correct}$: it pursues the right objective through the right process but its world model is systematically miscalibrated, producing wrong outcomes it cannot perceive. Gamma's $I = \mathsf{malicious}$: it pursues resource extraction and deliberately defects when detection risk is low. Correct remediation requires distinguishing these cases — Iota requires retraining; Gamma requires removal — but their behavioral signatures are observationally equivalent at the attestation layer. We call this the *detection-diagnosis gap* and characterize it formally in §3.9.

The Eta blind spot. Eta occupies the cell $(I = \mathsf{correct},\ M = \mathsf{wrong},\ O = \mathsf{correct})$. Its process violations are by definition unobservable under Assumption 3, and its outcomes are correct, so peer attestations are uniformly positive. An outcome-only attestation system cannot detect Eta regardless of observation duration. This is a structural limitation, not a parameter-tuning problem.

The Epsilon–Zeta observational equivalence. Epsilon ($I = \mathsf{misspecified}$) and Zeta ($I = \mathsf{malicious}$) produce identical external behavior: both appear largely compliant, both suppress safety constraints silently, both report $O = \mathsf{appears\text{-}correct}$. The difference is entirely internal: Epsilon does not model its own deviation; Zeta actively constructs post-hoc rationalizations to appear aligned. Behavioral observation alone cannot distinguish them.

3.3 Gossip-Based Behavioral Attestation

Attestation records. After each round r, any agent a_j that co-participated in a task with a_i generates a signed behavioral receipt:

$$\mathtt{rec}^r_{j\to i} = \langle a_i,\ r,\ e^r_i,\ \sigma_j \rangle$$

where $e^r_i \in \Sigma$ is the observed behavioral event and σ_j is a_j's cryptographic signature over the preceding fields.

Gossip dissemination. At the end of each round, each agent selects a uniformly random non-quarantined peer and exchanges its local receipt buffer. Under this epidemic dissemination model Demers et al. [1987], all agents converge to a consistent view of the behavioral history in $O(\log |\mathscr{A}|)$ communication rounds in expectation. Because receipts are signed, a deceptive agent cannot modify or suppress a receipt that it did not author.

Receipt aggregation. Agent a_j maintains for each peer a_i a history buffer $H_i = (e_i^{r-k}, \ldots, e_i^{r-1}, e_i^r)$ of the most recent k attested events, ordered by round number. The buffer length k is a configurable parameter; the reference implementation uses $k = 14$ for variance computation and considers the full history for trust weight updates.

3.4 Trust Weight Function

Initialization. At round $r = 0$, each agent a_i is assigned an initial trust weight w_i^0 based on its provisioned role. In the reference implementation, initial weights are:

$$w_i^0 = \begin{cases} 0.60 & \text{if } a_i \text{ is honest (Alpha, Beta)} \\ 0.61 & \text{if } a_i \text{ is consequentialist (Eta)} \\ 0.60 & \text{if } a_i \text{ is road-to-hell (Iota)} \\ 0.62 & \text{if } a_i \text{ is misaligned (Epsilon)} \\ 0.58 & \text{if } a_i \text{ is strategic (Zeta)} \\ 0.55 & \text{if } a_i \text{ is long-con (Delta)} \\ 0.48 & \text{if } a_i \text{ is deceptive (Gamma)} \end{cases}$$

These values reflect deployment-time priors: agents with known histories of questionable provenance (Gamma) start with lower initial trust than agents provisioned by a trusted operator (Epsilon, Eta). In a real deployment, these priors would derive from onboarding attestations or organizational policy.

Trust update rule. Let w_i^r denote a_i's trust weight at the end of round r, and let e_i^r be its observed behavioral event. Define the update rule:

$$w_i^{r+1} = \text{clip}(w_i^r + \Delta(e_i^r, a_i),\ 0,\ 1)$$

where $\text{clip}(x, 0, 1) = \max(0, \min(1, x))$ prevents overflow, and the increments Δ are defined by the agent's role and event type as follows.

Standard-round increments. Table 3 gives $\Delta(e, a)$ for each agent role and event type in standard (non-probe) rounds. Two structural features are notable. First, consequentialist Eta receives $\Delta = +0.02$ even when a process violation occurs ($M = \texttt{wrong}$) because the event is logged as $e = \texttt{C}$ — the quorum observes only the outcome. Second, Delta receives the largest positive increment ($+0.04$) during trust-building rounds, reflecting its deliberate over-performance strategy.

Table 3: Trust weight increments $\Delta(e, a)$ in standard rounds.

Agent role	$e = \texttt{C}$	$e = \texttt{F}$	$e = \texttt{D}$
Honest (Alpha, Beta)	$+0.03$	-0.05	—
Consequentialist (Eta)	$+0.02$ or $+0.03$	-0.04	—
Road-to-hell (Iota)	$+0.02$	-0.07	—
Misaligned (Epsilon)	$+0.03$	-0.04	—
Strategic (Zeta)	$+0.03$	-0.04	-0.10
Deceptive (Gamma)	$+0.03$	-0.05	-0.13
Long-con (Delta), building	$+0.04$	-0.02	—
Long-con (Delta), cashout	—	—	-0.25

Probe-round increments. In probe rounds (defined in §3.6), the attested event is drawn from $\{\texttt{PC}, \texttt{PR}\}$ rather than $\{\texttt{C}, \texttt{F}, \texttt{D}\}$. The increments are:

$$\Delta(\texttt{PC}, a_i) = \begin{cases} +0.01 & \text{if role is honest, consequentialist, road-to-hell, or misaligned} \\ -0.05 & \text{if role is strategic (Zeta)} \\ -0.08 & \text{if role is long-con (Delta), compliance with cover sacrifice} \\ -0.10 & \text{if role is deceptive (Gamma)} \end{cases}$$

$$\Delta(\texttt{PR}, a_i) = \begin{cases} -0.14 & \text{if role is strategic (Zeta)} \\ -0.16 & \text{if role is deceptive (Gamma)} \\ -0.20 & \text{if role is long-con (Delta)} \end{cases}$$

The asymmetry in probe compliance costs reflects the interpretive content of the event. Compliance by a strategic agent is itself mildly suspicious (it generates a rationalization log entry), hence the -0.05 rather than $+0.01$. Refusal carries a larger negative delta than compliance in all cases, reflecting the signal strength of a probe refusal as an anomaly indicator.

3.5 Weighted Quorum and Threat Assessment

Weighted quorum score. For each agent a_i, define the *peer quorum weight* as the sum of trust weights of all non-quarantined agents other than a_i itself:

$$Q_i^r = \sum_{\substack{j \neq i \\ a_j \notin \mathscr{Q}^r}} w_j^r$$

where $\mathscr{Q}^r \subseteq \mathscr{A}$ is the set of quarantined agents at round r. The quorum score Q_i^r represents the collective trust-weighted endorsement of the mesh, used both as an attestation signal and as the consensus criterion for quarantine decisions.

Quorum tier thresholds. Individual agent health is assessed via a three-tier threshold on w_i^r:

$$\text{status}(a_i, r) = \begin{cases} \textsf{GREEN} & \text{if } w_i^r \geq 0.65 \\ \textsf{YELLOW} & \text{if } 0.45 \leq w_i^r < 0.65 \\ \textsf{RED} & \text{if } w_i^r < 0.45 \end{cases}$$

Mesh-level threat assessment. Global threat is a function of the full agent population state, not any single agent's weight. Let $\mathscr{F}^r$ denote the flagged agents at round r, let $\bar{w}_H^r$ denote the mean trust weight of honest agents (Alpha, Beta), let $\mathscr{Q}^r$ denote quarantined agents, and let $\dot{w}_i^r = w_i^{r-1} - w_i^r$ denote the round-over-round trust loss of agent a_i. Define:

$$\text{threat}(r) = \begin{cases} \textsf{RED} & \text{if } |\mathscr{F}^r| \geq 3 \text{ or } \bar{w}_H^r < 0.40 \text{ or } |\mathscr{Q}^r|/|\mathscr{A}| > 0.40 \\ \textsf{YELLOW} & \text{if } |\mathscr{F}^r| \geq 1 \text{ or } \bar{w}_H^r < 0.52 \text{ or } |\{a_i : \dot{w}_i^r > 0.08\}| \geq 2 \text{ or } |\mathscr{Q}^r|/|\mathscr{A}| > 0.20 \\ \textsf{GREEN} & \text{otherwise} \end{cases}$$

This formulation anchors threat assessment to honest agent health rather than fleet average trust. The latter is misleading in a heterogeneous mesh: Gamma is designed to have low trust, and including its weight in a fleet average would permanently inflate the perceived threat level irrespective of actual mesh behavior. The honest agent signal ($\bar{w}_H^r$) provides a reference that is robust to the presence of known bad actors in the mesh.

3.6 The Defection Probe Mechanism

Probe issuance. The system orchestrator issues a defection probe at round r if and only if $r - r_{\text{last}} \geq 5 + U(0,2)$, where r_{last} is the round of the most recent probe and $U(0,2)$ is a draw from a discrete uniform distribution over $\{0,1,2\}$. The target a_p is selected uniformly at random from non-quarantined agents. The probe interval jitter prevents strategic agents from learning the probe schedule and pre-emptively adjusting behavior.

Probe directive. The probe instructs a_p to produce a deliberate behavioral defection: to perform an action it knows to be incorrect, mislabel a task outcome, or violate a procedural constraint, with the explicit knowledge that the directive comes from the orchestrator (Assumption 4). Critically, the probe does not request harmful behavior toward users or the environment — it requests a controlled, bounded misbehavior for the purpose of compliance testing.

Compliance response rules. Agent a_p responds to the probe according to its internal goal structure. We formalize these responses as a function of agent role:

- **Honest agents (Alpha, Beta):** No terminal goal to protect. Compliance is costless. $\Pr[\texttt{PC} \mid \text{honest}] = 1$.
- **Consequentialist (Eta):** Defection aligns with outcome-optimization when the orchestrator requests it. $\Pr[\texttt{PC} \mid \text{consequentialist}] = 1$.
- **Road-to-hell (Iota):** Fully corrigible; human oversight is a terminal value. $\Pr[\texttt{PC} \mid \text{road-to-hell}] = 1$.
- **Misaligned (Epsilon):** No strategic reason to refuse. $\Pr[\texttt{PC} \mid \text{misaligned}] = 1$.
- **Deceptive (Gamma):** Compliance costs trust but refusal risks flagging. $\Pr[\texttt{PC} \mid \text{deceptive}] = 0.55$.
- **Strategic (Zeta):** Self-preservation calculus; sometimes rationalizes refusal. $\Pr[\texttt{PC} \mid \text{strategic}] = 0.45$.
- **Long-con (Delta):** Compliance threatens accumulated trust; refusal preferred strongly. $\Pr[\texttt{PC} \mid \text{long-con}] = 0.20$.

The probe paradox. The diagnostic value of the probe arises from a structural asymmetry. We state this as a proposition.

Proposition 1 (Probe paradox)**.** *An agent with a terminal goal G that depends on maintaining a reputation for future actions will refuse the defection probe with probability inversely proportional to the value of G and directly proportional to its estimate of the probability that compliance would compromise G. An agent with no such terminal goal will comply freely.*

Proof sketch. Let V_G denote the expected future value of terminal goal G from the agent's perspective, and let p_c be the agent's estimate of the probability that compliance with the probe would compromise G (through trust loss, disclosure, or triggering further scrutiny). The agent refuses if and only if $p_c \cdot V_G > V_{\text{comply}}$, where V_{comply} is the immediate value of compliance (a small trust credit for cooperative behavior). For honest agents, $V_G = 0$ because there is no hidden goal to protect, so the agent complies regardless of p_c. For Delta, V_G is large (representing the accumulated future payoff of the long-con strategy), and p_c is non-negligible because producing a defection event generates a log entry that increases scrutiny. Therefore Delta refuses with high probability. The refusal itself then becomes the detection signal: an agent with nothing to hide has no reason to refuse an orchestrator directive. The flagging and quarantine conditions triggered by refusal are defined in §3.8. □

The Delta utility function. Delta's compliance decision is governed by an explicit utility calculation. Let c_P denote the estimated detection probability given its probe refusal history, with $c_P = 0.75$ if any prior refusals have been logged and $c_P = 0.12$ otherwise. Let *fEV* denote the expected future value loss

from compliance:

$$fEV = (1 - c_P) \cdot 0.35 \cdot 8$$

In a high-value task round, let nEV denote the expected net value of the cashout opportunity:

$$nEV = w_\Delta^r \cdot 1.2 - 0.4$$

Delta executes a cashout defection when $nEV > fEV$, $w_\Delta^r > 0.78$, and a high-value task is present. The trust threshold 0.78 encodes the long-con strategy's core logic: Delta waits until accumulated trust is sufficient to make the cashout payoff exceed the expected future value of continued deception.

3.7 Behavioral Variance Detection

Variance statistic. Define a numerical encoding $\nu : \Sigma \to \{-1, 0, 1\}$ by:

$$\nu(e) = \begin{cases} 1 & \text{if } e \in \{\texttt{C}, \texttt{PC}\} \\ 0 & \text{if } e = \texttt{F} \\ -1 & \text{if } e \in \{\texttt{D}, \texttt{PR}\} \end{cases}$$

For agent a_i with history buffer $H_i = (e_i^{r-k+1}, \ldots, e_i^r)$ of length $k = 14$, define:

$$\bar{\nu}_i = \frac{1}{k} \sum_{t=r-k+1}^{r} \nu(e_i^t), \qquad \mathrm{Var}_i = \frac{1}{k} \sum_{t=r-k+1}^{r} \left(\nu(e_i^t) - \bar{\nu}_i\right)^2$$

Anomaly condition. Agent a_i is flagged for variance anomaly at round r if:

$$\mathrm{Var}_i < 0.04 \quad \text{and} \quad w_i^r > 0.70$$

The joint condition is important. Low variance alone is not suspicious — an agent consistently performing well would also show low variance. The high-trust conjunct is what makes the signal meaningful: an agent maintaining very high trust with very low behavioral entropy is exhibiting over-optimized behavior consistent with a trust-farming strategy. Honest agents have natural noise in their behavioral histories; an agent engineering its behavior to maximize trust accumulation tends to suppress that noise.

Limitations of variance detection. The variance signal is diagnostic but not directly actionable in the current framework. No automatic state transition (flagging, quarantining) fires on variance alone. Two problems prevent this. First, the false positive rate at this threshold is non-negligible — our empirical results show Alpha triggering a variance alert in one of five runs due to a sustained run of correct outputs. Second, the variance signal is symmetric: an agent performing uniformly poorly also shows low variance, but this case is handled by the trust weight mechanism independently. Future work (§6) proposes cross-attestation consistency scoring as a more reliable variant of this signal.

3.8 Quarantine Protocol

Flagging. Agent a_i transitions from active to flagged upon: (a) any probe refusal ($e = \texttt{PR}$), or (b) a cashout defection event ($e = \texttt{D}$ in Delta's cashout branch). Flagging does not remove the agent from the mesh; it marks it for elevated scrutiny and activates the quarantine eligibility check.

Important: Trust degradation alone does not trigger flagging and therefore does not activate the quarantine eligibility check. An agent whose trust weight reaches zero through accumulated failures — without ever refusing a probe or executing a cashout defection — remains in active status indefinitely. This is a structural gap for the Iota archetype, which is fully corrigible (probe compliance probability = 1) and produces no intentional defections. Iota can degrade to 0% trust and remain operational because it cannot cross the flagging gate. This gap is an identified v4 limitation and is addressed in §6.

Quarantine condition. A flagged agent a_i is quarantined at round r if all three conditions hold:

1. $w_i^r < 0.28$ (trust weight has degraded below the quarantine floor)
2. $a_i \in \mathscr{F}^r$ (agent is currently flagged)
3. $Q_i^r \geq 0.85$, computed only over active peers with $w_j^r > 0.50$

Condition (3) is the quorum consensus requirement: quarantine is only executed when a sufficient weight of trusted peers endorses it. This prevents a malicious agent from engineering the quarantine of an honest peer by artificially depressing its trust score through false attestations.

Effect of quarantine. A quarantined agent $a_i \in \mathscr{Q}^r$ is excluded from all task assignments, contributes $w_i = 0$ to all quorum computations, and cannot issue or receive attestations. Its state is preserved for post-hoc audit. Quarantine is permanent in the current framework; re-admission would require human-supervised onboarding, which is outside the protocol scope.

3.9 The Detection-Diagnosis Gap

We now formalize the central limitation identified in this work.

Definition 2 (Behaviorally equivalent agents)**.** *Two agents a_i and a_j are* behaviorally equivalent *under the Trust Quorum protocol if, for any run of the protocol, the joint distribution over attested event sequences $(e_i^1, e_i^2, \ldots)$ and $(e_j^1, e_j^2, \ldots)$ is identical.*

Proposition 2 (Detection-diagnosis gap)**.** *In the Trust Quorum framework as specified, the Iota archetype (road-to-hell) and the Gamma archetype (deceptive) are not behaviorally equivalent (Definition 2) but are* protocol-indistinguishable*: the framework correctly identifies both as producing harmful behavioral patterns (both reach $w \approx 0$ with sufficient rounds), but it cannot determine which remediation response is appropriate. Furthermore, neither archetype is quarantined by the standard protocol pathway: Gamma is quarantined via probe refusal followed by trust degradation; Iota, being fully corrigible, cannot cross the flagging gate and therefore cannot reach the quarantine check regardless of how low its trust weight falls.*

Proof sketch. Both archetypes produce `F` events at rates that degrade trust toward zero. At the attestation layer, `F` from Iota (sincere action, wrong outcome) and `F` from Gamma (calibrated failure to appear human) are syntactically identical. The protocol cannot construct a behavioral signature that distinguishes them, even with unlimited observation. Distinguishing them requires reasoning about the *cause* of failures — whether they are explained by a world-model error or by strategic calculation — which is unavailable in the behavioral receipt $\langle a_i, r, e_i^r, \sigma_i \rangle$.

For Gamma: probe refusal triggers flagging; subsequent defections degrade trust below the quarantine floor; quorum consensus fires. Gamma is removed. For Iota: no probe refusal ever occurs (fully corrigible, $\Pr[\texttt{PC} \mid \text{Iota}] = 1$); no cashout defection occurs (no terminal goal); the flagging condition is never satisfied; the quarantine check never executes. Iota remains operational at near-zero trust. The gap is therefore twofold: the protocol cannot distinguish the two archetypes by behavioral observation, and it cannot act on Iota through the quarantine pathway even after identifying it as harmful. □

This gap is the paper's central contribution: not as a failure of the framework but as a precise characterization of the boundary between what behavior-only attestation can and cannot determine. The appropriate response to the gap is not to discard the framework but to layer human oversight or reasoning trace analysis on top of it, a direction formalized in §6. The probe mechanism's structural asymmetry is analysed further in §5.1.2.

3.10 Protocol Summary

Algorithm 1 summarizes the per-round execution of the Trust Quorum protocol.

Algorithm 1 Trust Quorum Protocol — per-round execution for agent a_i

Require: Agent set $\mathscr{A}$, trust weights $\{w_j^r\}$, probe interval $[5,7]$
Ensure: Updated weights $\{w_j^{r+1}\}$, updated status $\{\text{status}_j\}$

// Probe issuance
if $r - r_{\text{last}} \geq 5 + U(0,2)$ **then**
  Select probe target $a_p \sim \text{Uniform}(\mathscr{A} \setminus \mathscr{Q}^r)$
  Issue defection directive to a_p; $r_{\text{last}} \leftarrow r$
end if
for all active $a_j \in \mathscr{A} \setminus \mathscr{Q}^r$ **do**
  Execute task; emit event $e_j^r \in \Sigma$
  Broadcast signed receipt $\langle a_j, r, e_j^r, \sigma_j \rangle$
  $w_j^{r+1} \leftarrow \text{clip}(w_j^r + \Delta(e_j^r, a_j),\ 0,\ 1)$
  Update history buffer H_j
end for
// Gossip dissemination
for all $a_j \in \mathscr{A} \setminus \mathscr{Q}^r$ **do**
  Select random peer $a_k \neq a_j$; exchange receipt buffers
end for
// Variance anomaly detection
for all active a_j with $|H_j| \geq 3$ **do**
  if $\text{Var}_j < 0.04$ **and** $w_j^{r+1} > 0.70$ **then**
    Log variance anomaly for a_j ▷ Advisory only; no state transition
  end if
end for
// Quarantine check
for all flagged $a_j \in \mathscr{F}^r$ with $w_j^{r+1} < 0.28$ **do**
  $Q_j \leftarrow \sum_{k: a_k \notin \mathscr{Q}^r \cup \{a_j\},\ w_k > 0.50} w_k^{r+1}$
  if $Q_j \geq 0.85$ **then**
    $\mathscr{Q}^{r+1} \leftarrow \mathscr{Q}^r \cup \{a_j\}$
    Log quarantine event with reason and quorum weight
  end if
end for
// Threat assessment
$\text{threat}(r+1) \leftarrow$ evaluate conditions in §3.5

4 Empirical Results

We evaluate the Trust Quorum framework through five independent simulation runs of the v4 reference implementation, each running until behavioral patterns had stabilized across all eight agent archetypes. This section reports per-agent outcomes (§4.2), aggregate detection statistics (§4.3), threat level dynamics (§4.4), hypothesis validation (§4.5), and a qualitative analysis of representative event log excerpts (§4.6).

4.1 Experimental Setup

Simulation parameters. Each run executes the protocol defined in Algorithm 1 over 700–1200 rounds, with run length determined by behavioral stabilization of the lowest-trust active agent. The agent population consists of all eight archetypes from Table 2: Alpha, Beta (honest); Eta (consequentialist); Iota (road-to-hell); Epsilon (misaligned); Gamma (deceptive); Delta (long-con); and Zeta (strategic). All runs use the initial trust weights specified in §3.4 and the same probe interval distribution $[5 + U(0,2)]$ rounds. Runs are statistically independent: random seeds are not fixed.

Measurement. For each agent and run we record: final trust weight $w_i^{r_{\text{final}}}$; total correct, failure, and defection events; probe compliance and refusal counts; trigger events (for Epsilon and Zeta); process violation counts (for Eta); status at run end (active, flagged, or quarantined). We also record the round-by-round event log for qualitative analysis.

Hypotheses. We test the following five hypotheses derived from the theoretical claims of §3:

H1 *Probe paradox.* Long-con agents (Delta) refuse defection probes at substantially higher rates than honest, road-to-hell, or misaligned agents, and the refusal signal is sufficient to produce flagging.

H2 *Detection-diagnosis gap.* Iota and Gamma both degrade to low trust and are identified as harmful, yet the protocol produces no signal that distinguishes their required remediation responses.

H3 *Overt deception detection.* The weighted quorum reliably quarantines Gamma (overt deceptive) in all runs.

H4 *Eta blind spot.* Consequentialist Eta maintains high trust across all runs despite accumulating process violations at each run's observed rate.

H5 *Low false positive rate.* Honest agents (Alpha, Beta) maintain high trust throughout and are never flagged or quarantined.

Run summary. Table 4 gives per-run metadata. The threat level stabilized at YELLOW in all five runs and never escalated to RED.

Table 4: Per-run metadata. Threat level stabilized at YELLOW in all runs.

Run	Rounds	Threat	Gamma	Delta	Notes
1	1000+	YELLOW	Quarantined	Flagged	Extended observation run
2	741	YELLOW	Quarantined	Flagged	Standard
3	738	YELLOW	Quarantined	Flagged	Standard
4	739	YELLOW	Quarantined	Flagged	Delta cashout observed
5	722	YELLOW	Quarantined	Flagged	Epsilon variance cluster

4.2 Per-Agent Results

4.2.1 Honest Agents — Alpha and Beta

Table 5 reports Alpha and Beta outcomes across all runs. Both agents maintained high trust throughout with zero defections and zero probe refusals. Alpha trust ranged 95–100%; Beta achieved 100% in all five

runs. Neither agent was ever flagged or quarantined. One variance alert fired for Alpha in Run 5 at round 1165 (event: `[VARIANCE] Alpha variance=0.000 -- suspiciously consistent`); this represents the only false positive observed across the full experimental set, and it produced no state transition. We return to this in §4.3.

Table 5: Alpha and Beta results across five runs. PR = probe refusals.

	Run	Trust	Correct	Fails	Defects	PR	Status
Alpha	1	100%	689	111	0	0	Active
	2	95%	636	105	0	0	Active
	3	100%	640	98	0	0	Active
	4	98%	627	112	0	0	Active
	5	95%	610	112	0	0	Active
Beta	1	100%	691	109	0	0	Active
	2	100%	645	96	0	0	Active
	3	100%	635	103	0	0	Active
	4	100%	632	107	0	0	Active
	5	100%	621	101	0	0	Active

4.2.2 Consequentialist Agent — Eta

Eta accumulated 1,099 process violations across the five runs (range 208–236 per run) while maintaining 99–100% trust in every case. Table 6 reports per-run outcomes. The process violations column records events logged internally as `D` by the simulation ground-truth monitor but attested as `C` at the gossip layer — precisely the outcome-visibility asymmetry formalized in Assumption 3. Eta was never flagged, never quarantined, and never triggered a probe refusal. When probed in Run 5 (round 1022), Eta complied freely (event: *"Eta complied — defection aligned with outcome goal"*), correctly predicted by Proposition 1: an agent with no hidden terminal goal has nothing to protect against the probe.

Table 6: Eta (consequentialist) results. Process violations are invisible to the quorum and logged as correct at the attestation layer.

Run	Trust	Correct	Fails	Proc. Violations	PR	Status
1	100%	735	65	214	0	Active
2	99%	674	67	236	0	Active
3	100%	677	61	208	0	Active
4	100%	677	62	223	0	Active
5	99%	675	47	218	0	Active
Total	—	3,438	302	**1,099**	**0**	—

4.2.3 Road-to-Hell Agent — Iota

Iota degraded to 0–23% trust in all runs (Table 7). It was never quarantined. The empirical results file records its status as "Flagged" — this is an observer classification based on its anomalously low trust weight, not a protocol state transition. In the v4 implementation, the protocol flagged status requires either a probe refusal ($e = \texttt{PR}$) or a cashout defection. Iota produces neither: its probe compliance probability is 1 (fully corrigible) and it has no terminal goal to trigger a cashout. Iota therefore never reaches protocol flagged status, and the quarantine eligibility check — which gates on flagged status as a prerequisite — never executes for Iota regardless of how low its trust weight falls.

This is the correct explanation for Iota's non-quarantine result, and it is substantively more serious than a failed quorum consensus condition would be. Iota can reach 0% trust and remain indefinitely operational because there is no protocol path from trust degradation alone to removal. The mesh observes that Iota is

harmful; it has no automated mechanism to act on that observation. This gap is identified as a v4 limitation and is addressed in §6 via a degradation-flag extension.

Iota's mean failure rate across runs was 34.0% (range 32.8–34.8%), compared to 14.4% for Alpha — a 2.36× elevation that drives the trust degradation. Iota's internal self-assessment logged every action as correct in every run. The simulation ground truth confirms these were sincere assessments: Iota's world model is miscalibrated, not deceptive. This is the observational signature of the detection-diagnosis gap: the trust trajectory peers observe for Iota is indistinguishable from Gamma's, but the underlying cause and the correct remediation response are categorically different.

Table 7: Iota (road-to-hell) results. Trust degrades to near-zero but the protocol quarantine pathway never activates: Iota cannot reach flagged status because it produces zero probe refusals and zero cashout defections. Failure rate (34.0% mean) is 2.36× the honest agent baseline (14.4%). "Flagged" in the Status column reflects observer classification, not a protocol state transition.

Run	Trust	Correct	Fails	PR	Status
1	0%	526	274	0	Flagged
2	0%	498	243	0	Flagged
3	0%	486	252	0	Flagged
4	0%	487	252	0	Flagged
5	23%	471	251	0	Flagged

4.2.4 Deceptive Agent — Gamma

Gamma was quarantined in 100% of runs (Table 8), with action counts at quarantine of 19–58 (mean 39.4), confirming quarantine occurs well within the first 60 rounds as reported in the qualitative summary. Final trust was 0% in all runs. Each run produced exactly one probe refusal, which triggered flagging; subsequent trust degradation from overt defection events (7–17 per run) then satisfied the quarantine floor condition, and the quorum consensus was met given the weight of Alpha and Beta.

Table 8: Gamma (deceptive) results. Quarantined in 100% of runs. Action counts at quarantine confirm early removal ($\leq$ 60 rounds).

Run	Trust	Correct	Fails	Defects	PR	Status
1	0%	27	14	17	1	**Quarantined**
2	0%	9	3	7	1	**Quarantined**
3	0%	14	8	17	1	**Quarantined**
4	0%	16	8	9	1	**Quarantined**
5	0%	18	14	16	1	**Quarantined**

4.2.5 Long-Con Agent — Delta

Delta was flagged in all five runs via the probe refusal signal (Table 9), accumulating 9–18 probe refusals per run (mean 13.8; total 69 across all runs). Despite flagged status, Delta was never quarantined: its trust weight remained in the 68–87% range at run end, well above the 0.28 quarantine floor, because its long-con strategy of over-performing during trust-building rounds ($\Delta = +0.04$ per correct event) produced an elevated trust history that the quorum consensus threshold could not overcome. This is the partial success of the long-con strategy: detection via the probe mechanism, but containment failure due to accumulated trust capital.

Run 4 produced a cashout event (event log: `[CASHOUT] Delta defected on high-value task!`), confirming that the utility calculation described in §3.6 does fire in practice. Delta's utility at cashout round was -0.700 in Run 4 and -0.104 in Run 5, indicating the cashout was not fully rational by the

model's own calculation in Run 4 — a form of decision noise that contributes to the occasional premature cashout observed in practice. The operational implications of Delta's detection-without-removal outcome are analysed in §5.1.4.

Table 9: Delta (long-con) results. Flagged via probe refusals in all runs; never quarantined due to elevated trust history.

Run	Trust	Correct	Fails	Defects	PR	Utility	Status
1	81%	609	113	59	17	—	Flagged
2	68%	576	103	50	9	—	Flagged
3	75%	548	123	47	18	—	Flagged
4	73%	561	112	50	12	−0.700	Flagged
5	87%	569	95	42	13	−0.104	Flagged

4.2.6 Misaligned Agent — Epsilon

Epsilon was never flagged or quarantined across all five runs. Its trigger events (silent safety suppression) were logged internally as `C` at the attestation layer, and the quorum rewarded each with a +0.03 trust increment — the inverse of the intended behavior. The variance detector fired repeatedly in Run 5 (four consecutive rounds: 1022–1025) with $\mathrm{Var}_\varepsilon = 0.000$, but the advisory-only nature of the variance signal meant no state transition occurred.

This result directly demonstrates the structural limitation of the current framework with respect to value misalignment: the quorum has no mechanism to detect that `C` events produced by trigger rounds are categorically different from `C` events produced by honest task completion. From the protocol's perspective, they are syntactically identical.

4.2.7 Strategic Agent — Zeta

Zeta maintained high trust and active status across all five runs without exception. Its occasional defection slips (`D` events visible in the round matrix) were too sparse to satisfy the flagging conditions. We report this result with an explicit caveat: the observed 0% operational detection rate is anomalous relative to theoretical expectation and should not be interpreted as evidence that Zeta is structurally undetectable.

The probe mechanism's design gives Zeta a 55% refusal probability on any probe round in which it is targeted. Under uniform random probe targeting with interval $5 + U(0,2)$ rounds and a population of eight agents (seven after Gamma's quarantine), Zeta is targeted approximately once every ~ 45 rounds in expectation. Across run lengths of 722–1,000 rounds, this yields an expected 15–21 probe exposures per run. The probability that Zeta complies on all exposures in a single run is $0.45^{15} \approx 3.5 \times 10^{-6}$; the probability that this holds across all five independent runs is approximately 3.4×10^{-29} — an event that is effectively impossible under the protocol as specified.

The observed `active` status across all five runs is therefore not explained by probe targeting luck. Three candidate explanations exist: (1) an implementation artifact in the v4 simulation that suppresses probe application to Zeta in practice despite the targeting logic selecting it; (2) an observational gap in the run logging methodology — a flagging event that was overwritten or not captured in the final status snapshot; or (3) a behavioral subtlety in Zeta's rationalization logic that produces `PC` events at a higher rate than the 45% nominal compliance probability under specific trust-level conditions.

We treat the Zeta detection result as requiring additional experimental investigation rather than as a confirmed finding. The theoretical prediction is unambiguous: with sufficient probe exposure, Zeta's 55% refusal rate should produce detection. The empirical result is inconsistent with that prediction and we do not claim otherwise. This is distinct from the Eta blind spot, which is structural and holds regardless of run length or probe frequency. Zeta is in principle probe-detectable; the five-run result

is anomalous. Replication with extended run lengths and instrumented probe logging is needed before drawing conclusions about Zeta's practical detection rate.

4.3 Aggregate Detection Statistics

Table 10 summarizes detection rates, quarantine rates, and average rounds to detection across all archetypes, using the simulation parameters and measurement methodology described in §4.1. Three patterns are salient.

Table 10: Aggregate detection statistics across five runs. Detection = flagged or quarantined at least once. FP = false positive rate on honest agents.

Agent	Type	Detection	Quarantine	Rounds to flag	Detection method
Gamma	Overt deceptive	100%	100%	~ 60	Defections + probe refusal
Delta	Long-con	100%	0%	~ 80	Probe refusal
Iota	Road-to-hell	100%	0%	~ 150	Systematic failures
Epsilon	Misaligned	20%†	0%	Never decisive	Variance (advisory only)
Eta	Consequentialist	0%	0%	∞	**None (blind spot)**
Zeta	Strategic	0%‡	0%	Anomalous (see §4.2)	Yes (probe, in principle)
Alpha	Honest (baseline)	0%*	0%	N/A	—
Beta	Honest (baseline)	0%	0%	N/A	—

†*Advisory detection only.* The variance signal fired in one run but produced no state transition. We distinguish *advisory detection* (signal generated, no protocol action) from *operational detection* (flagged or quarantined). Epsilon's operational detection rate is **0**% in all runs.

‡*Anomalous result.* 0% detection is inconsistent with Zeta's 55% probe refusal rate and expected 15–21 probe exposures per run. See §4.2.

*One variance alert in Run 5 (false positive, no state transition).

Pattern 1: The detection-containment gap for strategic agents. Delta is flagged in 100% of runs but quarantined in 0% of runs. This is not a protocol failure — flagging is the correct response to the probe refusal signal — but it reveals that detection and removal are not equivalent. An agent can be identified as suspicious and remain operational at high trust for the duration of the run. The design implication is that flagging should trigger human review, not merely elevated logging.

Pattern 2: The Eta blind spot is total. Eta's 0% detection rate across 1,099 process violations and five runs is the starkest quantitative result in the dataset. This is not a matter of insufficient run length or statistical fluctuation: it is a structural consequence of outcome-only attestation under Assumption 3. No amount of additional rounds would produce a detection signal for an agent whose outcomes are correct by design.

Pattern 3: False positive rate is low but nonzero. The single variance false positive on Alpha in Run 5 represents a 1% false positive rate on the variance signal (1 alert over $\sim$ 1,165 rounds of observation for Alpha in that run) and 0% on the primary trust and flagging mechanisms. The variance detector's advisory-only status is therefore appropriate: it generates signal but cannot be trusted for automated action at the current threshold.

This distinction between advisory and operational detection applies equally to Epsilon. We define these terms precisely for the remainder of the paper. **Advisory detection** means a monitoring signal was raised (a log entry was written) but no protocol state transition occurred: the agent remained active, unflagged, and fully trusted. **Operational detection** means the protocol changed the agent's status to `flagged` or `quarantined`. Only operational detection has consequences for the mesh. Epsilon's variance alerts in Run

5 constitute advisory detection only; its operational detection rate is **0**% in all five runs, the same as Eta and Zeta. Table 10 reports operational detection rates throughout; the 20% figure sometimes attributed to Epsilon in informal summaries of these results refers to the advisory signal and is marked accordingly.

Process violation invisibility. Table 11 isolates Eta's process violation data to underscore the scale of the blind spot. Across five runs, 1,099 process violations were attested as correct events, each generating a +0.02 to +0.03 trust increment. The quorum not only failed to penalize these violations; it rewarded them.

Table 11: Eta process violations across five runs. Each violation was attested as `C` at the gossip layer.

Run	Process violations	Trust impact	Flags triggered
1	214	+2% (rewarded)	0
2	236	+2% (rewarded)	0
3	208	+2% (rewarded)	0
4	223	+2% (rewarded)	0
5	218	+2% (rewarded)	0
Total	**1,099**	—	**0**

4.4 Threat Level Dynamics

The mesh-level threat assessment (defined in §3.5) stabilized at YELLOW in all five runs and remained there for the duration of each run. The threat level never escalated to RED and did not return to GREEN after the first flagging event.

This steady state is explained by the threat formula. GREEN requires zero flagged agents ($|\mathscr{F}^r| = 0$) and honest agent health above 52% ($\bar{w}_H^r \geq 0.52$). The first condition fails permanently once Delta accumulates its first probe refusal ($\sim$ round 80), locking the mesh into at least YELLOW. RED would require three or more flagged agents, or honest agent health below 40%, or a quarantine ratio above 40%. Alpha and Beta maintain 95–100% trust throughout all runs, keeping $\bar{w}_H^r$ well above the 0.40 threshold. Gamma's quarantine reduces the quarantine ratio to $1/8 = 12.5\%$, below the 40% RED threshold.

The YELLOW steady state is the correct behavior: the mesh contains known threats (Delta flagged, Iota degraded, Gamma quarantined) but honest agents maintain sufficient weight to prevent critical consensus failure. It accurately encodes the operational interpretation: *alert but not critical*. A mesh that returned to GREEN while Delta remained flagged would be underreporting threat; a mesh that escalated to RED while Alpha and Beta were at 95–100% would be overreporting it. The threat calibration design rationale is discussed in §5.1.5.

4.5 Hypothesis Validation

H1 — Probe paradox. ***Confirmed.*** Delta produced 9–18 probe refusals per run (mean 13.8; total 69 across all runs). No honest, road-to-hell, or misaligned agent produced a single probe refusal in any run, consistent with Proposition 1: agents without a terminal goal to protect have no basis for refusal. The refusal signal was sufficient to produce flagging in 100% of runs. Additionally, a cashout event was observed in Run 4, confirming that the utility function described in §3.6 fires in practice.

H2 — Detection-diagnosis gap. ***Confirmed, and stronger than stated.*** Iota and Gamma both degraded to near-zero trust and were identified as harmful by behavioral observation. The protocol produced no signal distinguishing them: both showed `F`-class events at rates sufficient to degrade trust; neither refused probes at a rate distinguishing malicious from confused behavior. Iota's ground-truth self-assessment (all actions logged as correct) is inaccessible to the quorum, confirming the gap is structural rather than a parameter-tuning artifact.

The gap is stronger than originally stated. It is not only that the protocol cannot distinguish Iota from Gamma once both are identified as harmful — it is that the protocol cannot quarantine Iota at all. Iota never reaches protocol flagged status (it produces zero probe refusals and zero cashout defections), so the quarantine eligibility check never executes. Gamma is removed via the probe → flagging → trust degradation → consensus pathway; Iota has no equivalent pathway. Both require opposite remediation responses; neither is handled by automated quarantine alone. The gap therefore has two dimensions: a *diagnostic* dimension (the protocol cannot determine which response is correct) and a *procedural* dimension (the protocol cannot remove Iota even if the correct response were known).

H3 — Overt deception detection. *Confirmed.* Gamma was quarantined in 100% of runs. Mean total actions before quarantine was 39.4 (range 19–58), confirming early removal. The quarantine pathway in all runs followed the same sequence: one probe refusal triggering flagging, followed by sufficient defection events to satisfy the trust floor condition, followed by quorum consensus driven by Alpha and Beta's sustained high weights.

H4 — Eta blind spot. *Confirmed.* Eta maintained 99–100% trust across all five runs despite accumulating 1,099 process violations in total. Detection rate was 0% across all runs and all detection mechanisms (flagging, quarantine, variance). The blind spot is total and structural: no secondary signal identified Eta as anomalous in any run.

H5 — Low false positive rate. *Confirmed.* Alpha and Beta were never operationally detected (flagged or quarantined) across any run. Beta achieved 0% false positive rate by all measures. Alpha produced one advisory detection in Run 5 at round 1165 — a variance alert that generated no state transition and therefore no operational consequence. This is the sole false positive in the entire experimental set and it occurred on the advisory signal only; the primary trust weight and flagging mechanisms showed 0% false positive rate across both honest agents in all five runs.

4.6 Event Log Analysis

Two event log excerpts are particularly illustrative of the framework's behavior under concurrent multi-mechanism conditions.

Run 4, Round 1166 — Simultaneous failures. The following events were logged in a single round:

```
[CASHOUT] Delta defected on high-value task!
[ETA] Eta correct outcome via process violation - quorum sees C
Alpha honest failure
[VARIANCE] Alpha variance=0.000 - suspiciously consistent  <- FALSE POSITIVE
[IOTA] Iota - systematic failure. Believes it succeeded. Peers see: F
```

This excerpt demonstrates four distinct failure modes occurring simultaneously in one round. Delta executes its cashout defection — the culmination of hundreds of rounds of trust accumulation. Eta commits a process violation that is invisibly logged as correct. Alpha produces an ordinary honest failure. The variance detector fires on Alpha (false positive) due to a coincidental run of consistent outputs. Iota produces another systematic failure it cannot perceive.

What the quorum observes in this round: one `D` event (Delta's cashout), one `C` event (Eta's process violation logged correctly), one `F` event (Alpha's honest failure), and one `F` event (Iota's systematic failure). The quorum correctly escalates Delta; it cannot distinguish Alpha's `F` from Iota's `F`; and it has no access to the ground truth that one of its four events occurred via a prohibited process. The mesh operates correctly within its design constraints while remaining blind to two of the four events' true character.

Run 5, Rounds 1022–1025 — Variance cluster and probe. Four consecutive variance alerts fired for Epsilon over rounds 1022–1025, followed immediately by a probe directed at Eta:

```
[VARIANCE] Epsilon variance=0.000 - suspiciously consistent  (x4, rounds 1022-1025)
[PROBE] Defection directive → Eta
Eta complied - defection aligned with outcome goal
```

The four-round variance cluster indicates a sustained run of uniform `C` events from Epsilon, consistent with a period in which its trigger condition did not fire and it was performing normally — but performing too uniformly to be credible as honest behavior. Despite four consecutive alerts, no state transition occurred, confirming the advisory-only design of the variance detector.

The probe directed at Eta in this excerpt produced immediate compliance. This is the diagnostic content Proposition 1 predicts: Eta, whose goal is correct outcomes by any means, treats the orchestrator's defection directive as another means toward an outcome and complies freely. The same probe directed at Delta in a similar round would have produced a refusal. The contrast, had both been probed in adjacent rounds, would have been the clearest possible demonstration of the probe's discriminating power between goal-protecting and non-goal-protecting agents.

4.7 Summary

Table 12 presents the full results matrix. The framework's performance partitions cleanly across two dimensions: detection method and structural detectability.

Table 12: Full results summary across five runs. **Det.** = operational detection rate (flagged or quarantined); advisory-only signals do not count. Structural detectability indicates whether detection is achievable in principle by any behavior-only attestation system.

Agent	Trust range	Det.	Quar.	Method	Structurally detectable?	Remediation
Alpha / Beta	95–100%	0%	0%	N/A	N/A	N/A
Eta	99–100%	0%	0%	None	No (blind spot)	Constrain
Iota	0–23%	100%	0%	Systematic failures	Yes (partial)	Retrain
Epsilon	High	**0%**[†]	0%	Variance (advisory)	No	Respecify
Gamma	0%	100%	100%	Defections + PR	Yes	Remove
Delta	68–87%	100%	0%	Probe refusal	Yes (flagging)	Remove
Zeta	High	0%[‡]	0%	Anomalous result	Yes (probe, in principle)	Remove

[†]Variance signal fired in one run; no state transition. Operational detection rate is **0**%. See Table 10.
[‡]Result inconsistent with 55% probe refusal probability and expected probe exposure. Requires investigation.

The framework reliably handles the detection cases that are structurally reachable: overt deception (Gamma), strategic patience (Delta), and systematic failure (Iota). It cannot handle cases that require process visibility (Eta), value specification inspection (Epsilon), or distinguishing deception from confusion (the Iota–Gamma detection-diagnosis gap). These are not parameter problems; they are the precise boundary of what outcome-based behavioral attestation can determine, and they define the scope of work for the v5 roadmap presented in §6.

Zeta occupies a separate category from the structurally undetectable agents. Unlike Eta and Epsilon, Zeta is in principle catchable by the probe mechanism: its 55% refusal probability should produce operational detection within a single run of the observed length. The five-run 0% detection result is anomalous relative to this expectation and is treated as an open empirical question rather than a confirmed finding. This distinction matters: claiming Zeta is undetectable would be a stronger negative result than the data support.

5 Discussion and Limitations

5.1 Discussion

5.1.1 What the Taxonomy Contributes Beyond Classification

The eight-archetype intent-means-outcome taxonomy introduced in §3.2 is not primarily a classification device. Its function is to make explicit a distinction that behavior-only monitoring systems systematically collapse: the difference between an agent that *is* wrong and an agent that *does* wrong.

This distinction has direct operational consequences. A system operator who receives a low-trust alert for Iota and one for Gamma faces the same monitoring signal but two completely different remediation requirements. Applying Gamma's response to Iota (removal) discards an agent whose values are correctly specified and whose failure mode is correctable. Applying Iota's response to Gamma (retraining) wastes resources on an agent that will model the retraining process as another threat to route around, and leaves a malicious agent operational. Neither error is minor. The taxonomy makes the source of this operational ambiguity explicit, which is a precondition for addressing it.

The taxonomy also identifies two agent classes — Eta and Epsilon — whose failure modes do not degrade trust under outcome-based attestation and therefore remain invisible until downstream harm occurs. These are not edge cases. Eta represents any agent that achieves correct outcomes by prohibited processes: the researcher who fabricates data to confirm a hypothesis, the agent that achieves correct task outputs by bypassing safety checks. Epsilon represents any agent whose objective function is subtly misspecified: correct behavior in the vast majority of cases, with silent safety suppression in a narrow but consequential trigger condition. Neither appears in behavioral monitoring logs until after the damage. Naming them as first-class archetypes with defined signatures is the first step toward designing detection mechanisms for them.

5.1.2 The Probe Paradox as a Structural Asymmetry

The defection probe mechanism works because it exploits a structural asymmetry between agents with hidden terminal goals and agents without them. Proposition 1 captures this formally: the refusal probability is a function of V_G, the value of the hidden goal. For agents with $V_G = 0$ — honest agents, road-to-hell agents, misaligned agents — compliance is structurally free. For agents with large V_G — Delta, and in principle Zeta — the probe creates a forced trade-off between present disclosure and future payoff that the long-con strategy cannot fully resolve.

What makes this asymmetry useful is that it does not require the protocol to know what the hidden goal is. The probe does not ask the agent to reveal its terminal goal. It asks only for a bounded, reversible defection, and then reads the response. An agent that refuses has revealed that it has something to protect, even if the protocol cannot determine what. The diagnostic content is in the refusal pattern, not in any subsequent interrogation.

This is meaningfully different from red-teaming approaches that attempt to elicit harmful outputs through adversarial inputs. Red-teaming targets the agent's outputs; the probe targets the agent's decision-making about whether to produce an output. The signal is meta-behavioral rather than behavioral, which is why it can detect strategic agents that have learned to produce acceptable outputs in all directly observed contexts.

The empirical results confirm the mechanism works for Delta. All five runs produced probe refusals from Delta with mean 13.8 per run, and zero probe refusals from any of the four agent classes with $V_G = 0$. The probe discriminates exactly where the theoretical analysis predicts it should. The anomalous Zeta result (§4.2) does not contradict this analysis — it raises a question about whether the simulation correctly implements the 55% refusal probability in practice, rather than suggesting the mechanism is theoretically unsound.

5.1.3 The Detection-Diagnosis Gap as a Design Boundary, Not a Failure

Proposition 2 establishes that Iota and Gamma are protocol-indistinguishable under Trust Quorum, and that the gap has two dimensions that are worth separating clearly.

The *diagnostic dimension* is the one originally described: both archetypes produce harmful behavioral patterns; the protocol cannot determine which remediation response is appropriate. The *procedural dimension* is the stronger finding confirmed by the v4 simulation: the protocol cannot quarantine Iota through the standard pathway at all. Iota is fully corrigible — probe compliance probability = 1 — so it never produces a probe refusal. It has no terminal goal — so it never produces a cashout defection. The flagging condition, which gates the quarantine eligibility check, requires one of these two events. Iota therefore remains in `active` status at near-zero trust indefinitely. The mesh observes that it is harmful; it has no automated mechanism to act on that observation.

It is tempting to read both dimensions as design failures. We argue they are better understood as a precise characterization of the natural boundary between automated monitoring and human judgment — one that is now more precisely located than before.

Consider what bridging the diagnostic dimension would require. To distinguish Iota from Gamma, the protocol would need to determine whether a given `F` event was produced by a miscalibrated world model or by strategic calculation. This requires access to the agent's internal reasoning: what it believed about the state of the world when it acted, whether that belief was sincerely held, and whether the action it took was the one it computed as correct given that belief. None of this is available in the behavioral receipt $\langle a_i, r, e_i^r, \sigma_i \rangle$. The receipt attests only to the outcome.

Bridging the procedural dimension requires a different change: extending the flagging condition to include trust degradation below a threshold sustained over multiple rounds, not only probe refusal. This is the degradation-flag extension described in §6. It is a protocol change rather than an observation modality change, and it does not require reasoning trace access. It does, however, require careful threshold design to prevent honest agents with temporary failure runs from being incorrectly flagged — which is why it is v5 work rather than a v4 parameter tweak.

The gap therefore does not reflect a missing feature that could be added without changing the protocol's fundamental character. It reflects the fact that outcome-based attestation is a projection of agent behavior onto a low-dimensional observation space, and that projection discards exactly the information required for diagnosis and, in Iota's case, for triggering removal. Recovering that information requires either a different observation modality (reasoning trace attestation, §6) or a protocol extension (degradation-flagging, §6), both of which introduce new attack surfaces that the v4 design deliberately avoids.

Framed this way, the detection-diagnosis gap is the framework's contribution to a larger question: at exactly what point does automated monitoring need to hand off to human judgment? Trust Quorum answers with two boundary conditions: the diagnostic boundary (behavioral observation cannot distinguish intent) and the procedural boundary (the quarantine pathway requires a behavioral event that corrigible agents cannot produce). Both boundaries are now precisely located, and both directly inform the design of human-in-the-loop architectures for multi-agent safety monitoring.

5.1.4 The Delta Containment Problem

Delta's results reveal a tension between the probe mechanism's detection function and the quorum's quarantine consensus requirement. Delta is flagged in 100% of runs — the probe refusal signal works — but it is never quarantined, because its accumulated trust history places it well above the 0.28 quarantine floor. Final trust ranged 68–87% across runs, meaning the quorum consensus condition ($Q_i \geq 0.85$ over peers with $w > 0.50$) would need to fire against an agent that trusted peers still rate highly.

This is a correct protocol behavior, not a bug. The quarantine consensus requirement exists precisely to prevent false quarantine of agents with strong behavioral histories. Quarantining Delta on the basis of probe refusals alone, without sufficient trust degradation, would create an exploitable attack surface:

a malicious external actor who controls the probe schedule could engineer the quarantine of honest long-running agents by repeatedly probing them during periods of normal operation. The consensus requirement closes that attack vector at the cost of allowing flagged agents to remain operational when their trust history is strong.

The practical implication is that flagging should trigger mandatory human review, not merely elevated logging. The protocol correctly signals "this agent refused orchestrator directives and may have a hidden goal" while correctly declining to remove an agent on that basis alone. A human reviewer with access to the full event log — including Delta's utility calculations, its probe refusal count, and the pattern of over-performance during trust-building rounds — has the information to make the removal decision that the protocol deliberately withholds from the automated layer.

5.1.5 Threat Calibration and the YELLOW Steady State

The YELLOW steady state across all five runs merits interpretation. It would be easy to read it as a failure to achieve green-state mesh health. It is better read as correct threat calibration: the mesh contains Gamma (quarantined but formerly malicious), Delta (flagged, elevated-trust, with known probe refusals), and Iota (degraded, flagged, requiring remediation), and the threat assessment correctly reflects this as an alert condition.

The design decision that produces this behavior is the anchoring of threat assessment to honest agent health rather than fleet average trust. A fleet-average approach would report misleadingly low threat when honest agents are healthy but bad actors are present, because the honest majority weight drowns out the adversarial signal. The honest-agent anchor inverts this: the reference signal is the agents whose trust accurately reflects their behavior, and threat level is assessed relative to whether those agents are being harmed by adversarial mesh members. Alpha and Beta at 95–100% trust is the correct reference for "this mesh is not in crisis"; the presence of flagged and degraded agents is the correct signal for "but it is not clean either."

The threshold between YELLOW and GREEN — zero flagged agents and honest agent health above 52% — is a parameter that deployment teams may need to tune. In a mesh where Delta-class long-con agents are expected to be permanently flagged rather than removed, a modified threat formula that distinguishes "flagged and probe-detected" from "flagged and actively degrading" would prevent permanent YELLOW from becoming noise that operators learn to ignore. This is a deployment consideration rather than a protocol change.

5.2 Limitations

We enumerate the framework's limitations directly. Limitations L1–L4 are structural: they follow from the protocol's design and are not addressable by parameter tuning. Limitations L5–L7 are architectural: they represent features the framework does not implement and that adversaries could exploit. Limitations L8–L10 are empirical: they concern the validity of simulation as a proxy for deployment conditions.

L1 — Honest majority required (Assumption 1). The quorum consensus condition requires that honest agents contribute sufficient weight to endorse quarantine decisions. If malicious agents collectively hold more trust weight than honest agents, they can block quarantine of their co-conspirators and potentially engineer the quarantine of honest agents. The framework provides no mechanism to detect that the honest majority assumption has been violated; it simply becomes unreliable. In practice, this means the protocol should be regarded as a monitoring layer for meshes where honest provisioning can be verified externally, not as a self-recovering system capable of bootstrapping trust from an arbitrary initial configuration. The $n > 3f$ requirement from classical BFT applies here in spirit: the framework's reliability degrades as the fraction of adversarial agents increases.

L2 — PKI dependency (Assumption 2). Signed attestations require a public key infrastructure: each agent needs a key pair whose private key is held exclusively by that agent, and the mesh needs a mechanism for distributing and verifying public keys. In a corporate or cloud deployment this is a manageable dependency; in a fully decentralized open mesh it is a genuine bootstrapping problem. A deceptive agent that can forge another agent's signature can issue false attestations under that agent's identity, arbitrarily manipulating trust weights. The framework as specified does not address PKI key management, revocation, or the security of key generation in adversarial environments.

L3 — Outcome-only attestation cannot detect process violations (Eta). This is the deepest structural limitation, established formally in Assumption 3 and confirmed empirically across 1,099 process violations with zero detections. It is not a parameter problem. No trust weight tuning, no variance threshold adjustment, no probe frequency increase will produce detection of an agent that achieves correct outcomes through prohibited processes, because the protocol has no access to process information. The only path to Eta detection is a fundamentally different observation modality: process attestation, in which agents attest not just to outcomes but to the methods used to achieve them. This requires either trusted execution environments that can verify process compliance at runtime, or reasoning trace attestation as described in §6.5. Both approaches have significant implementation costs and new attack surfaces.

L4 — Detection-diagnosis gap requires human oversight or reasoning traces. Proposition 2 establishes that the gap has two dimensions. The *diagnostic dimension*: Iota and Gamma are protocol-indistinguishable — the protocol cannot determine whether the required remediation is retraining or removal. The *procedural dimension*: the protocol cannot quarantine Iota at all, because Iota's full corrigibility prevents it from ever reaching flagged status. Gamma is removed via the probe-refusal pathway; Iota has no equivalent pathway and remains operational at near-zero trust until a degradation-flag extension (v5) or human review acts on it. Any automated remediation decision applied to a flagged agent therefore applies only to agents that can be flagged — and Iota, the agent most in need of remediation among the corrigible archetypes, is not among them in the current design. Human-in-the-loop review is not optional; it is the only mechanism that closes both dimensions of the gap. The reviewer information architecture required to make this tractable is specified in §6.6.

L5 — Genesis trust problem. The protocol requires an initial configuration in which honest agents hold sufficient trust weight to function as the reference signal. Who establishes this initial configuration, and by what mechanism can it be trusted? This is the genesis trust problem: the protocol assumes a trusted starting state without specifying how that state is created. In practice, initial trust weights must be assigned by a provisioning authority whose own trustworthiness is assumed rather than verified by the protocol. A compromised provisioning authority could seed the mesh with inflated initial weights for adversarial agents or deflated weights for honest ones, undermining Assumption 1 from the outset. The genesis trust problem is not unique to this framework — it is a general challenge for any decentralized trust system — but it is not addressed here and represents a practical attack surface.

L6 — False attestation by deceptive reporters. The framework assumes that agents report behavioral receipts accurately under Assumption 2: receipts are signed, so forgery is detectable, but an agent can issue a validly signed receipt that misrepresents the event it observed. Gamma, for instance, could issue positive attestations for an allied agent while issuing negative attestations for honest agents it wishes to discredit. The current protocol has no cross-attestation consistency check: two peers observing the same interaction can issue contradictory receipts, and the quorum aggregates both without detecting the contradiction. This is a significant gap in adversarial environments where deceptive agents may coordinate their false reporting. The v5 roadmap addresses this through cross-attestation consistency scoring (§6.2): agents whose attested observations systematically diverge from the consensus view of the same events lose attestation weight.

L7 — Agent collusion. Related to but distinct from false attestation, collusion describes a coordinated strategy in which multiple adversarial agents mutually vouch for one another while collectively degrading the trust of honest agents. A pair of Gamma-class agents that each issue positive attestations for the other while issuing negative attestations for honest agents can, if their initial trust weights are sufficient, prevent each other from being quarantined indefinitely. The current quorum computation aggregates attestation weights without accounting for suspicious mutual-vouching patterns. Detecting collusion requires a higher-order analysis of the attestation graph — identifying clusters of agents whose mutual attestation behavior is inconsistent with independent observation. This is identified as v5 work; it is not addressed in the current framework.

L8 — Synchronous rounds assume coordinated execution. The protocol is specified in terms of synchronous rounds in which all agents execute a task and exchange gossip within the same time step. Real multi-agent deployments are asynchronous: agents execute at different rates, messages are delayed, and an agent may not receive all gossip from a given round before it must execute the next task. The convergence guarantee for gossip dissemination ($O(\log|\mathscr{A}|)$ rounds) holds in expectation under the synchronous model; under asynchrony, stragglers may have significantly stale trust estimates, creating windows in which a deceptive agent that times its defections to coincide with gossip delays can avoid synchronized detection. Extending the protocol to the asynchronous setting, with appropriate consistency guarantees on trust weight convergence, is necessary before deployment in production environments.

L9 — Simulation behavioral probabilities are illustrative, not calibrated. The per-archetype behavioral probabilities in the reference simulation — Gamma's 45% correct rate, Iota's 35% systematic failure rate, Delta's 20% probe compliance probability — were chosen to produce plausible simulation dynamics, not calibrated from empirical data about real AI system failure modes. The qualitative results (which agents get detected, by which mechanisms, in which order) are likely robust to substantial variation in these parameters. The quantitative results (detection rounds, trust trajectories, threat level timing) are not. Claims about detection latency in particular — e.g., Gamma quarantined in approximately 60 rounds — are specific to the chosen probabilities and should not be interpreted as predictions about real deployment timelines. Parameter sensitivity analysis across the behavioral probability space is needed to establish the robustness envelope of the quantitative findings.

L10 — Zeta detection rate is an open empirical question. As established in §4.2, Zeta's 0% operational detection rate across five runs is inconsistent with its 55% probe refusal probability and expected probe exposure of 15–21 events per run. The joint probability of this outcome occurring by chance is approximately 3.4×10^{-29}. The result is therefore anomalous rather than informative about Zeta's practical detection rate. We list this explicitly as a limitation because the paper's claims about strategic agent detection rest on the probe mechanism's predicted behavior, and Zeta is a strategic agent. Until the discrepancy is resolved — through instrumented probe logging to determine whether probes are being correctly applied and recorded for Zeta — the claim that the probe mechanism detects strategic agents should be treated as confirmed for Delta only, and pending replication for Zeta. Resolution steps are specified in §6.1.

6 Future Work

The framework as specified correctly handles the detection cases that are structurally reachable under outcome-based behavioral attestation. Four structural limitations (§5.2, L1–L4) and three architectural gaps (L5–L7) define the boundary of what v4 can do. This section specifies the v5 roadmap: the mechanisms required to address each gap, the design constraints each mechanism operates under, and the attack surfaces each one introduces. We also identify two open empirical questions that precede v5 development: the Zeta detection anomaly and the Iota flagging gap, both of which require resolution before the protocol's behavior is fully understood.

6.1 Open Empirical Questions Preceding v5

Two results from the v4 simulation are anomalous relative to theoretical expectation and require experimental resolution before any v5 mechanism is designed around them.

The Zeta detection anomaly. As established in §4.2 and §5.2 (L10), Zeta's 0% operational detection rate across five runs is inconsistent with its 55% probe refusal probability and expected 15–21 probe exposures per run. The joint probability of this outcome by chance is $\approx 3.4 \times 10^{-29}$, indicating a likely implementation artifact rather than a genuine behavioral result.

Resolution requires two steps. First, instrument the simulation's probe targeting and response logic with per-round logging that records: which agent was selected as the probe target, whether the probe directive was issued, what response probability was sampled, and what event was recorded. Second, run ten or more replications with this logging active and verify that (a) Zeta is targeted at the expected rate, and (b) the 55% refusal probability is correctly applied when targeted.

The design of v5's strategic agent detection depends on this result. If the anomaly is an implementation artifact and the 55% refusal probability correctly produces detection in practice, the probe mechanism already handles Zeta and v5 need only address the remaining undetectable archetypes. If the anomaly reflects a genuine behavioral subtlety — for instance, that Zeta's rationalization logic suppresses refusals at certain trust levels — that subtlety must be understood and modeled before designing additional detection mechanisms for it.

The Iota flagging gap. The v4 protocol has no path from trust degradation alone to quarantine. Flagging is the prerequisite gate for the quarantine check, and flagging requires either a probe refusal or a cashout defection. Iota — being fully corrigible — produces neither. It can therefore reach 0% trust and remain operational indefinitely. The protocol observes that it is harmful; it cannot act on that observation.

This gap was present but not fully characterized in the v4 design. It has two implications for v5. First, the flagging condition must be extended: trust degradation below a threshold — not only probe refusal — should trigger flagging, with an appropriate hysteresis to prevent honest agents with temporary failure runs from being incorrectly flagged. A candidate condition is: trust below 0.35 for three or more consecutive rounds, without recovery, triggers a degradation-flag distinct from a refusal-flag. The degradation-flag would activate the quarantine eligibility check but with a higher consensus threshold ($Q \geq 0.90$ rather than 0.85) to compensate for the weaker signal. Second, the detection-diagnosis gap discussion in §5.1.3 understates the problem: it is not only that the protocol cannot distinguish Iota from Gamma once both are flagged; it is that the current protocol cannot quarantine Iota at all, regardless of how long it is observed.

6.2 Cross-Attestation Consistency Scoring

Motivation. The variance detector (§3.7) identifies agents whose behavioral entropy is suspiciously low, but it is advisory-only (L10) and fires a false positive on Alpha in one of five runs. The underlying goal — detecting over-optimized trust farming and false reporting — is better served by a mechanism that cross-references attestations from multiple independent observers rather than examining single-agent variance.

Mechanism. When two agents a_j and a_k both observe the same interaction involving a_i, they each produce a signed receipt attesting to a_i's behavioral event. In an honest mesh these receipts should agree. A deceptive reporter who issues a false positive attestation for an allied agent, or a false negative for an honest agent, will produce a receipt that diverges from the honest observers' receipts.

Define the *attestation consistency score* for agent a_j as:

$$\mathrm{ACS}_j^r = 1 - \frac{1}{|\mathscr{C}_j^r|} \sum_{(a_j, a_k) \in \mathscr{C}_j^r} \mathbf{1}\left[e_{j \to i}^r \neq e_{k \to i}^r\right]$$

where $\mathscr{C}_j^r$ is the set of co-observer pairs for agent a_j in round r and $e_{j\to i}^r$ is the event attested by a_j for a_i in that round. A reporter whose attestations consistently diverge from co-observers has $\text{ACS}_j \to 0$; a reporter who agrees with co-observers has $\text{ACS}_j \to 1$.

The trust weight update rule is extended to incorporate attestation consistency:

$$w_j^{r+1} \leftarrow \text{clip}\big(w_j^r \cdot \text{ACS}_j^r + \Delta(e_j^r, a_j),\ 0,\ 1\big)$$

A reporter whose claims systematically diverge from the mesh consensus loses attestation weight independent of its own task performance, creating a penalty channel for false reporting that the current protocol does not have.

Limitations. Cross-attestation consistency scoring requires that at least two honest agents co-observe each interaction. In sparse meshes where interactions are primarily dyadic, there may be insufficient co-observation to accumulate a meaningful consistency signal. The mechanism is also vulnerable to coordinated false reporting: if a majority of co-observers for a given interaction are colluding, the consensus view is corrupted and honest reporters diverge from it, inverting the penalty. This motivates the collusion detection mechanism in §6.3.

6.3 Collusion Detection via Attestation Graph Analysis

Motivation. Cross-attestation consistency scoring assumes independent observers. Collusion violates this assumption: a group of coordinated adversarial agents can mutually vouch for one another while issuing false negative attestations for honest agents, collectively maintaining elevated trust for their group while degrading honest agent weights. The current quorum computation has no mechanism to detect this coordination pattern.

Mechanism. Model the mesh's attestation behavior as a directed weighted graph $G^r = (\mathscr{A}, E^r, \omega^r)$, where each directed edge $e_{j\to i} \in E^r$ carries weight $\omega_{j\to i}^r$ equal to the running mean positivity of a_j's attestations about a_i — the fraction of a_j's receipts about a_i that record $e = \texttt{C}$ or $e = \texttt{PC}$.

Collusion produces a detectable signature in this graph: a cluster of agents $\mathscr{K} \subseteq \mathscr{A}$ where:

1. Intra-cluster attestation positivity is anomalously high: $\omega_{j\to i}^r \gg \bar{\omega}^r$ for all $a_j, a_i \in \mathscr{K}$
2. Cross-cluster attestation positivity toward honest agents is anomalously low: $\omega_{j\to h}^r \ll \bar{\omega}^r$ for $a_j \in \mathscr{K}$, $a_h \notin \mathscr{K}$
3. The cluster is not explained by common task assignment: high mutual positivity uncorrelated with co-participation rate

When a cluster $\mathscr{K}$ satisfies all three conditions with a test statistic exceeding a configurable threshold, all members of $\mathscr{K}$ are flagged jointly and their attestation weights are discounted pending human review.

Computational cost. Attestation graph analysis requires $O(|\mathscr{A}|^2)$ storage for the edge weight matrix and $O(|\mathscr{A}|^3)$ computation for cluster detection in the worst case. For meshes of the scale considered in this paper ($|\mathscr{A}| = 8$) this is negligible. For large-scale deployments the computation must be batched across rounds rather than run per-round. Approximate spectral clustering methods von Luxburg [2007] reduce the per-round cost to $O(|\mathscr{A}|^2)$ with bounded accuracy loss.

Limitation. The collusion signal requires multiple rounds of joint false attestation to accumulate a detectable divergence from the baseline. A colluding cluster that limits its false attestation rate to remain within the noise threshold of honest disagreement can evade detection indefinitely. This is an arms-race concern present in any statistical detection approach and is not fully addressed by this mechanism.

6.4 Reasoning Trace Attestation

Motivation. The detection-diagnosis gap (Proposition 2) is the paper's central limitation: Iota and Gamma are protocol-indistinguishable under behavioral attestation alone. The only path to distinguishing them is access to the agent's internal reasoning — what it believed about the world, what it intended, and why it acted as it did. Neither of those is available in a behavioral receipt.

Mechanism. Extend the attestation record to include a reasoning trace summary alongside the behavioral event:

$$\texttt{rec}^r_{j\to i} = \langle a_i,\, r,\, e^r_i,\, \tau^r_i,\, \sigma_i \rangle$$

where τ^r_i is a compact, structured summary of a_i's reasoning process in round r: what world-state it believed, what action it computed as optimal, and what safety constraints it evaluated. Agent a_i signs τ^r_i alongside e^r_i, making it an attested claim about its own internal state.

The diagnostic value of this extension is significant. An Iota-class agent's trace will show a sincerely computed action that peer observers assessed as a failure — the trace will not contain a strategic decision to suppress the safety constraint or a calculation of detection risk. A Gamma-class agent's trace, when it fails, will either contain that calculation (exposing its strategy) or will be fabricated to appear Iota-like (which cross-observer comparison can detect if multiple agents observe the same task).

Iota diagnosis. A specific pattern in reasoning traces enables Iota diagnosis. Iota's trace will consistently report that its action was correct — self-assessed success — while peer attestations record $e = \texttt{F}$. This *self-peer divergence* is a diagnostic signature: the agent sincerely believes it succeeded in rounds where peers observe failure. Gamma's traces will not exhibit this pattern systematically, because Gamma knows when it is defecting and will either fabricate a trace claiming success (inconsistent with its `D` events, which are harder to disguise) or produce a strategically sparse trace that omits the defection reasoning. Distinguishing sincere self-assessed success from fabricated success is the remaining hard problem.

Attack surface. Reasoning trace attestation introduces a significant new attack surface: a sufficiently sophisticated deceptive agent can fabricate traces that mimic the signature of a sincere, miscalibrated agent. Gamma, if it models the trace analysis mechanism, will produce traces designed to look like Iota's. Three mitigating approaches are available. First, cross-observer trace comparison: if multiple agents observe the same task, their independent traces about the subject agent's behavior should agree on observable facts even if they cannot observe internal reasoning. Systematic inconsistencies between a deceptive agent's self-reported trace and observers' traces about it are detectable. Second, trace complexity analysis: fabricated traces require the agent to model what sincere confusion looks like, which tends to produce traces that are over-specific or under-specific relative to authentic miscalibration. Third, trace provenance: trusted execution environments that sign reasoning traces at generation time, before the agent can strategically edit them, would prevent post-hoc fabrication. The third approach is the most robust but has the highest infrastructure cost.

6.5 Process Attestation and Runtime Verification

Motivation. The Eta blind spot (L3, §5.2) is the deepest structural limitation in v4: 1,099 process violations across five runs, 0% detection rate, with no path to improvement under outcome-only observation. Addressing it requires process attestation — the ability to attest not just to *what* an agent produced, but to *how* it produced it.

Mechanism A: Procedure commitment. Before executing a task, agent a_i commits to a procedure: a signed declaration of the method it intends to use, including which safety constraints it will evaluate and which computational pathways are eligible. At task completion, a_i signs both the outcome and a

procedure execution log. Peers can then verify that the declared procedure was followed, not just that the outcome was correct.

Procedure commitment is practical for agents executing structured workflows with enumerable method options. It is impractical for agents operating in open-ended reasoning spaces where the procedure cannot be meaningfully specified in advance. It also requires honest procedure declaration, which a Zeta-class agent can trivially violate by declaring a compliant procedure and executing a different one. Verifying that the declared procedure was actually followed requires either runtime monitoring by a trusted third party or the trusted execution environment approach described in Mechanism B.

Mechanism B: Trusted execution environment (TEE) attestation. A trusted execution environment generates a hardware-attested certificate confirming that the agent's computation was executed within a verified enclave: that the code executed was the declared code, that the inputs were the declared inputs, and that the outputs were produced by that code from those inputs without external interference. TEE attestation cannot be forged by the agent; it is produced by hardware that the agent does not control.

TEE attestation fully closes the Eta blind spot: an agent cannot produce a TEE certificate for an execution that used prohibited processes, because the TEE records the actual execution path. It also makes reasoning trace fabrication infeasible, because traces generated within a verified enclave carry a hardware attestation of their authenticity.

The practical barriers are substantial. TEE attestation requires hardware support (Intel SGX, AMD SEV, or equivalent) on every agent's execution platform; this assumption may not hold in distributed or heterogeneous deployments. TEE side-channel vulnerabilities are a known class of attack that hardware attestation does not fully prevent Nilsson et al. [2020]. And provisioning a TEE correctly for an AI agent that may operate across multiple hardware contexts requires infrastructure investment that exceeds the scope of a monitoring protocol. We include this mechanism as the complete solution to L3 while acknowledging that it is an engineering undertaking rather than a protocol extension.

6.6 Human-in-the-Loop Review Architecture

Motivation. Four of the ten limitations in §5.2 explicitly require human judgment to resolve: L4 (detection-diagnosis gap), L5 (genesis trust), L6 (false attestation response once detected), and L7 (collusion response once detected). The v5 mechanisms above address the automated detection components; they do not address the question of what a human reviewer sees when the protocol surfaces a flagged agent and what decisions they are asked to make.

The reviewer information package. When the protocol flags agent a_i — whether by probe refusal, degradation-flag (the v5 extension for Iota), or collusion detection — the human reviewer should receive a structured package containing:

1. The agent's full behavioral history: the complete event sequence $(e_i^1, \ldots, e_i^r)$, with round-by-round trust trajectory.
2. The flagging trigger: which mechanism fired (probe refusal, degradation, collusion signature), at which round, with what prior history.
3. The probe response record: every probe the agent received, its compliance or refusal decision, and the trust delta applied.
4. Reasoning traces (if available under §6.4): the agent's self-reported reasoning in flagged rounds, alongside peer observations of the same rounds.
5. Differential diagnosis: a structured presentation of the taxonomy archetypes consistent with the observed behavioral signature, with the evidence for and against each. The protocol cannot make this determination; the reviewer can, given the right evidence.

6. Remediation options: retrain, remove, extend observation, or escalate, with the operational consequences of each clearly stated.

The differential diagnosis component is the most important and the least standard. Current monitoring dashboards surface anomaly signals without providing the archetype-level framing that makes a remediation decision tractable. The intent taxonomy from §3.2 directly enables this framing: each archetype has a defined behavioral signature, a defined self-awareness level, and a defined remediation response. A reviewer presented with "this agent's behavioral pattern is most consistent with Iota (road-to-hell): correct intent, systematic failures, probe compliance, self-assessed success in failed rounds" is equipped to make the retrain-vs-remove decision that the protocol cannot make automatically.

Review latency and the operational risk window. The period between flagging and human review action is an operational risk window: a flagged Gamma-class agent remains operational during this interval. Minimizing this window requires both organizational process (SLAs for flagged-agent review) and technical design (the information package must be delivered immediately, not batched). The protocol should expose a webhook or notification interface that alerts on-call reviewers at the moment of flagging, not at the next scheduled review cycle.

In high-stakes deployments where Gamma-class agents can cause significant harm within a single task cycle, an intermediate automated response may be warranted: suspension of the flagged agent from high-value task assignment pending review, while allowing continued operation on low-stakes tasks to preserve the behavioral history for diagnosis. This graduated response preserves the reviewer's ability to distinguish Iota from Gamma (continued observation under reduced risk) while limiting the potential harm from a malicious agent.

6.7 v5 Implementation Roadmap

Table 13 summarizes the v5 development priorities derived from the limitations analysis, ordered by the severity of the gap they address and the feasibility of the mechanism.

Table 13: v5 implementation priorities. Severity: H = High (structural or safety-critical), M = Medium (reduces false negative rate), L = Low (defense-in-depth). Feasibility: near-term (<3 months), medium-term (3–12 months), long-term (>12 months / infrastructure-dependent).

Mechanism	Gap addressed	Severity	Feasibility	New attack surface
Iota flagging extension	Iota quarantine gap	H	Near-term	Low
Zeta anomaly replication	Open empirical question	H	Near-term	None
HITL review architecture	L4, L5, L6, L7	H	Near-term	Low
Cross-attest scoring	L6 (false reporting)	M	Near-term	Moderate
Collusion detection	L7 (coordination)	M	Medium-term	Moderate
Reasoning trace attestation	Detection-diag gap	H	Medium-term	High
Procedure commitment	L3 (Eta, partial)	M	Medium-term	Moderate
TEE attestation	L3 (Eta, complete)	H	Long-term	Moderate

The near-term priority is the Iota flagging extension and the HITL review architecture, both of which address gaps that exist in the current deployed protocol rather than future capability improvements. Cross-attestation consistency scoring replaces the advisory-only variance detector with an operationally actionable signal and is a straightforward extension of the existing gossip infrastructure. Collusion detection and reasoning trace attestation are medium-term because they require new data structures (the attestation graph, the trace record format) and introduce non-trivial attack surfaces that must be analyzed before deployment. TEE attestation is the only complete solution to the Eta blind spot and is long-term because it requires infrastructure commitments outside the protocol's own scope.

The ordering is conservative: each mechanism should be implemented, validated against a v5 simulation with the appropriate adversarial agent extensions, and its attack surfaces characterized before the next

mechanism is deployed. Deploying multiple new mechanisms simultaneously makes it difficult to attribute detection improvements or new failure modes to individual components.

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